

to T1 relaxation. This is due to the fact that another RF pulse is applied just before the signal is received by the coil. Therefore the more relaxation that has occurred, the stronger the signal will be when flipped (nutated) by 90 degrees. Therefore, the RF signal that is measured after a 90 degree spin flip increases as the time between successive applied pulses is increased.

The influence of relaxation parameters on the MR signal is one of the central principles of tissue contrast in MRI. Any measurement technique that exploits differences in MR signal among volume elements (voxels) of the sample enhances contrast in the image. Just as contrast in computed tomography is influenced by factors such as electron density (no. of electrons per gram) and physical density (grams per cubic centimeter), contrast in MRI is influenced by differences in the relaxation parameters T1 and T2, and in the nuclear spin density, N(H) (number of spins per cubic centimeter). Measurement techniques that exploit differences in T1, T2, and N(H) among tissues are described in the next chapter.

RELAXATION TIMES (T1 AND T2) FOR BIOLOGIC MATERIALS

Biologic materials may be characterized to some degree by their T1 and T2 values (Table 23-2). However, there are several difficulties. For example, the exact values are not unique to composite substances. A bulk sample may contain materials having a range of values for T1 and T2. The temperature of a sample also influences relaxation.

The rate of interaction among spins and their surroundings determines the rate at which the spins in a higher energy state dissipate energy to their surroundings. Molecules in a sample are in constant motion and rotate with frequencies that range from zero up to a maximum value determined by the temperature. Any magnetic moment is influenced by the rotation of nearby magnetic moments. Spins may change energy states from parallel to antiparallel alignment (T1), and they may also move out of phase with other spins (T2) as a result of these interactions. A number of other interactions, the discussion of which is beyond the scope of this text, also play a role.^{7,17}

Longitudinal relaxation T1 has been shown to vary with magnetic field strength. Since magnetic field strength, along with the gyromagnetic ratio, determines the resonance frequency for a nucleus, T1 may be expressed as a function of resonance frequency. For a wide range of tissues, T1 for hydrogen can be approximated⁸ as

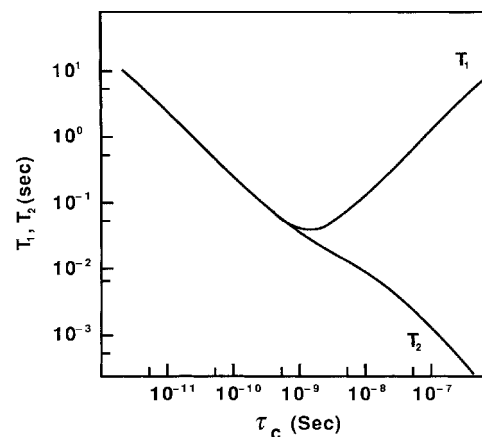
$$T1 = \alpha f^\beta \quad (23-6)$$

where T1 is in milliseconds, f is the resonance frequency in hertz, and the parameters α and β vary with tissue type over a range of $\alpha = 0.5$ to 10 and $\beta = 0.2$ to 0.4. The transverse relaxation time T2 has been found to be independent of resonance frequency (magnetic field strength).

The properties of a material that give rise to observed values of T1 and T2 are complex. However, a theory of the contribution of molecular motion to relaxation was suggested by Bloembergen et al. in 1948.¹⁷ An important characteristic of this theory is the correlation time of molecules. Molecules in liquids and semisolids are free to rotate. The rate at which they rotate is the rotational frequency. The inverse of the rotational frequency is the rotational period, also called the correlation time T_c (Margin Figure 23-12). Molecules that rotate more slowly have a greater probability of interacting with their neighbors. In such materials, relaxation can occur more rapidly, and the relaxation time constant is smaller. For T2 relaxation, the prediction is straightforward; a longer correlation time implies a faster transverse relaxation (shorter T2). For T1, however, there is a resonance phenomenon to consider. Molecules rotating at Larmor frequencies maximize their rate of interaction with their neighbors, just as the interaction of radio waves with magnetic moments is maximized at the Larmor

The mathematical equations that describe relaxation of the MR signal have the same general form as the equations that describe attenuation of photons from an x-ray beam as it passes through a material, and the decay of activity of a radioactive sample over time. In these processes, a "constant" characterizes the behavior of photons of specific energies in specific materials (attenuation coefficient, μ) or of certain types of atoms (decay constant, λ). In magnetic resonance, the constants, $1/T1$ and $1/T2$, characterize relaxation phenomena of specific nuclides.

In 1971, Damadian used NMR to observe differences in magnetic properties between normal and cancerous tissues in rats.⁹ These studies were extended to human tissue samples in 1973¹⁰ and stimulated considerable interest in the use of NMR as a biochemical probe. Lauterbur proposed the first imaging technique in 1973 and named it zeugmatography, from the Greek *zeugma* for "that which joins together." In that year, Lauterbur published the first MR images.¹¹ In 1974, an image of a mouse was obtained by Mallard.¹² The first human anatomic images were published by several groups in 1977 and included a wrist by Hinshaw et al.,¹³ a finger and various tissue phantoms by Mansfield and Maudsley,¹⁴ and an in vivo scan of the human thorax by Damadian and colleagues.¹⁵ The first MR image of the brain was obtained in 1980 at Nottingham by Holland et al.¹⁶



MARGIN FIGURE 23-12

Relaxation times T_1 and T_2 as a function of correlation time. Data for static magnetic field strength 1.4 tesla (proton resonance frequency, 60 MHz). (Redrawn from Martin, L., Delpuich, J. J., and Martin, G. J. *Practical NMR Spectroscopy*. London, Heydon, 1980.)

frequency. Thus a material having a correlation time that happens to coincide with the inverse of the Larmor frequency being used for MRI yields the minimum T1. Materials having smaller or larger values of correlation time have a larger T1. Furthermore, T1 depends on the Larmor frequency and increases as the magnetic field strength is increased. Thus, materials with correlation times near the resonance value for one field strength would have a short T1 at that field strength only. T2 is relatively unaffected by changes in resonance frequency and is therefore independent of field strength.

Correlation time analysis yields some straightforward predictions for T1 and T2 in biologic systems. Thermal motion is relatively fast in pure liquids such as water. Therefore, the correlation time is usually below the resonance value, and T1 is greater in water than in more viscous materials. Similarly, T1 in solids or anhydrous proteins is generally longer because the thermal motion is decreased and the correlation time is too long for resonance interactions. Because of the inverse relationship between T2 and correlation time, liquids tend to have longer T2s, and solids tend to have shorter T2s.

PROBLEMS

- 23-1. Show that the range of static magnetic field strengths 0.1 to 2.0 tesla corresponds to the resonance frequencies 4.3 to 86 MHz for hydrogen. By using data from Table 23-1, find the corresponding resonance frequencies for ^{19}F and ^{13}C .
- *23-2 Find the number of ^{13}C nuclei in 1 mg of pure ^{13}C . By using the data in Table 23-1, determine the number of milligrams of ^{13}C needed to give the same signal as a cubic millimeter of water (see Example 23-2).
- *23-3 A tissue T1 is characterized in Eq. (23-6) by $\alpha = 0.5$ and $\beta = 0.4$. What magnetic field strength is required to double the T1 for hydrogen that is obtained at a magnetic field strength of 0.35 tesla?
- *23-4 For a material having a T2 of 80 msec 320 msec after a 90° pulse, what is the relative magnitude of the MR signal compared to the maximum signal.

*For those problems marked with an asterisk, answers are provided on p. 493.

SUMMARY

- Nuclei with magnetic moments will precess in a static magnetic field. This precession may be monitored by the signal it induces in a receiver coil.
- Larmor equation: $f = \gamma B$.
- Quantum mechanical equation: $E = hf$.
- T1 is also known as longitudinal or “spin–lattice” relaxation.
- T2 is also known as transverse or “spin–spin” relaxation.
- The MR signal decreases in time after all RF pulsing has ceased, due to T1 and T2 relaxation.
- Relaxation properties of materials are determined by their correlation times, the period of rotation of the molecules.

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■ OBJECTIVES

After completing this chapter, the reader should be able to:

- Explain the phenomenon of free induction decay.
- State the rationale for use of the spin-echo technique in imaging.
- Define $T2^*$.
- Explain the pulsing and signal acquisition scheme used in the following pulse sequences:
 - Spin-echo
 - Carr–Purcell–Meiboom–Gill
 - Inversion recovery
 - Gradient-echo
- Chart the order of occurrence of RF pulses, gradient magnetic fields, and signal acquisition in spatially encoded spin-echo imaging.
- Explain how the two-dimensional Fourier transform (2DFT) method is used to construct the MR image.
- Calculate the time required to obtain an image using the spin-echo technique.
- List some motion suppression techniques.
- Give the physical principle that underlies the use of contrast agents in MR.
- Explain the physiological basis of functional MRI (fMRI).
- List reasons for tissue contrast, differences in MR signal among voxels, and how these differences are highlighted through the choice of pulse sequence parameters.
- Explain the chemical basis of MR spectroscopy and give some examples of nuclei that are studied.

■ OVERVIEW: MAGNETIC RESONANCE AS A PROBE OF THE BODY

The magnetic properties of nuclei described in Chapter 2 have significant applications in medical imaging and biochemical analysis. These applications are possible because the relaxation properties and the resonance frequency for a nucleus depend upon its environment. Factors such as the presence of chemical bonds, paramagnetic ions, and the rate of flow of fluids influence the magnetic resonance (MR) signal. Therefore, different regions of a biological sample produce different MR signals. While the reasons for these differences are often so complex that interpretation is difficult, their existence yields an intrinsic subject contrast between tissues. If the values of the MR signals can be mapped according to their spatial locations and can be encoded as brightness (gray scale) on a monitor, then an image may be produced. This is the method of magnetic resonance imaging (MRI). A plot of the magnitude of the MR signal as a function of frequency yields an MR spectrum. This latter approach is termed *magnetic resonance spectroscopy* (MRS).

■ PULSE SEQUENCES

Both imaging and spectroscopy require the application of radiofrequency (RF) pulses to a sample so that coherent signals may be obtained from a receiver coil placed in close proximity to the sample. These pulses are applied in specific sequences to produce MR signals that yield information about the sample. All “pulse sequences” have in common the use of RF pulses to tip (“nutate”) the bulk magnetization vector at some angle with respect to the static magnetic field.

The symbols B and H are used to indicate the presence of a magnetic field. While, for many purposes, they are used interchangeably, there are some differences that are important in the field of electromagnetism. H is the “magnetic field intensity.” It may be thought of as an externally applied “magnetizing force” much as a “gravitational” force is said to act upon objects in gravitational field theory. B is the “magnetic induction field.” It is the magnetic field that is actually induced in matter when that particular matter is placed in a given magnetic field intensity (H). In MR, we usually refer to the static (constant) magnetic field at the center of the MR scanner as the B_0 field (pronounced “bee-zero”).

Free Induction Decay

The simplest pulse sequence is free induction decay (FID) (Margin Figure 24-1). In FID, the magnetization is tipped (nutated) 90 degrees to orient the plane of precession of the protons perpendicular to the receiver coil. Such a pulse also has the effect of establishing “phase coherence” in the sample. That is, the nuclei are made to rotate together in phase. The signal is induced in the coil immediately after the 90-degree pulse. If another FID is performed, the magnitude of the resulting signal depends upon the amount of relaxation that has taken place since the last pulse.

There are two problems with using the FID sequence for imaging. First, it is difficult to record the MR signal immediately after the 90-degree pulse is transmitted. The transmitted pulse is orders of magnitude more intense than the received pulse. The presence of the large “transmitted” pulse induces a “ringing” in the receiver coil that requires at least a few milliseconds to dissipate. During this “ring-down” period the receiver coil is unable to respond to the signal induced by the rotating nuclei.

The second problem with the FID sequence is that the T2 relaxation of the tissue of interest is usually masked by another phenomenon that has the same effect upon the signal. The resonance frequency of a proton depends on the magnetic field to which it is exposed. Magnetic field inhomogeneities across the sample, particularly in imaging devices where the bore size is on the order of 100 cm, produce variations in the frequency of precession within the sample volume. These variations are indistinguishable from the “true” T2 signal decay produced as the protons interact with one another to go “out of phase.” The reduction in magnitude of the FID is shown in Margin Figure 24-1 as the result of two processes:

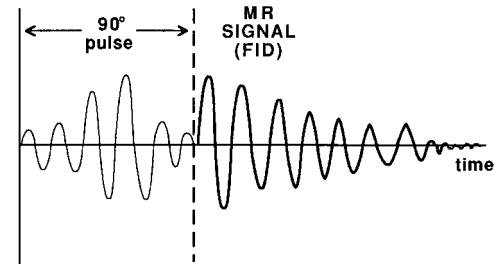
- Dephasing produced by magnetic field inhomogeneities governed by a time constant T2* (pronounced “T two star”)
- Dephasing produced by spin–spin interactions, governed by a time constant T2

Because the inhomogeneity phenomenon is a property of the MR system and not of the sample, we usually prefer to remove the T2* component of FID. The influence of magnetic field inhomogeneities may be reduced by applying a “rephasing pulse” described below in the spin-echo sequence.

Spin Echo

As in FID, the spin-echo pulse sequence (Margin Figure 24-2) begins with an RF pulse that nutates the magnetization to some angle (not necessarily 90 degrees). After application of the pulse, the individual spins are in phase, or “coherent.” With time, coherence (and therefore the MR signal) is reduced because of interactions among spins (“true” T2) and magnetic field inhomogeneities (T2*). The inhomogeneities may be as small as a few parts per million; nevertheless, the fact that the magnetic field is a little stronger in some regions of the sample than in others causes the nuclei to precess at slightly different rates and therefore to dephase.

To remove the effect of magnetic field inhomogeneities, a 180-degree “rephasing” pulse can be applied at some time T after the initial RF pulse. If the initial RF pulse produced nutation about the x-axis, the rephasing pulse is applied to produce nutation about the y-axis (Margin Figure 24-3). The effect of the rephasing pulse is to take spins that were precessing more rapidly and flip them behind the spins rotating at the average rate of precession. Similarly, spins that are rotating more slowly are automatically flipped “ahead” of the mean precession rate. The nuclei are still precessing at different rates because of inhomogeneities in the magnetic field of the system. After the rephasing pulse, the faster spins catch up with the average spins, and the slower spins are caught by the average spins. When this happens, the spins are said to have “rephased.”

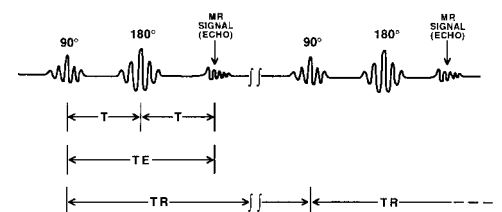


MARGIN FIGURE 24-1

The MR signal immediately following a 90-degree pulse decays rapidly due to the effects of magnetic field inhomogeneity. The MR signal for free induction decay (FID) is not shown to scale in this illustration. Note also that the 90-degree pulse and the received signal may occur in separate coils.

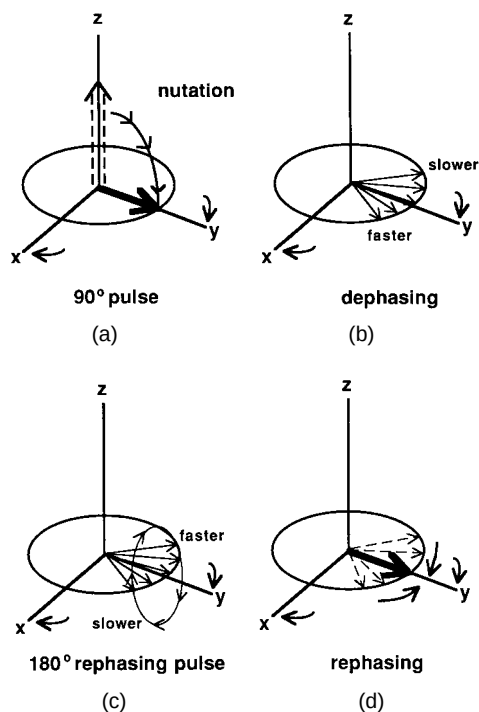
T2 is the dephasing of neighboring spins due to spin–spin interaction. The slightly different resonance frequencies among neighboring spins due to chemical shift, motion, and other effects causes the spins to lose phase coherence (stop precessing as a group with all spins pointing in the same direction).

T2* (pronounced “T two star”) is dephasing that is produced by inhomogeneities of the static magnetic field within a sample. This effect results in a loss of phase coherence that mimics the “natural” or “true” spin–spin relaxation of a sample. Even in the best of MR system magnets the value of the T2* time constant for the magnet is much smaller than the value of T2 for most features of interest. Therefore, if uncorrected, the sample would relax (lose MR signal) so rapidly from T2* relaxation that the small differences in true T2 that produce contrast in the image would be completely overshadowed by T2*.

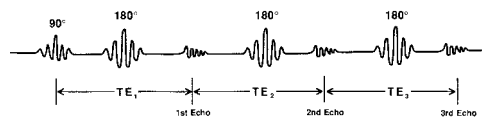


MARGIN FIGURE 24-2

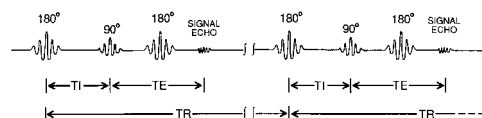
The spin-echo pulse sequence.

**MARGIN FIGURE 24-3**

In the spin-echo pulse sequence the 90-degree pulse (A) causes nutation of the bulk magnetization about the x-axis. Dephasing then occurs (B) as slower components of the magnetization lag and faster components advance. A 180-degree rephasing pulse (C) causes nutation about the y-axis and reverses the order of fast and slow components. The components then merge (D) to produce the maximum signal. Diagrams A to D are shown in the rotating frame of reference.

**MARGIN FIGURE 24-4**

The Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence.

**MARGIN FIGURE 24-5**

The inversion recovery (IR) pulse sequence.

If dephasing progresses for a time T after the initial RF pulse, then a time T is required after application of the rephasing pulse for the spins to rephase. At the moment of exact rephasing, the signal will be strongest, since the effect of T_2^* is completely removed. The time after the initial RF pulse that the moment of exact rephasing occurs is referred to as TE (time to echo) and is equal to $2T$. Thus we choose TE by choosing T , the time between the application of the 90-degree pulse and the 180-degree pulse. The entire sequence is repeated at a time TR (repetition time) after the first 90-degree pulse. Typically, TR is at least 10 times TE. Obviously, TR must exceed TE.

Carr-Purcell-Meiboom-Gill

Carr and Purcell introduced the idea of using multiple rephasing pulses, with collection of the signal after each pulse.¹ Meiboom and Gill proposed applying nutation about an axis that is perpendicular to the axis of nutation for the original RF pulse (as described above for spin echo).² These techniques are used in the Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence. Because the spins are rephased after TE, it is as if the experiment had started over with the initial 90-degree pulse. Therefore a second rephasing pulse delivered a time TE after the first produces a second rephasing, and another signal ("echo") is obtained. The diminution of signal between subsequent echoes is caused by T_1 and T_2 relaxation. The CPMG pulse sequence is diagrammed in Margin Figure 24-4. In the CPMG pulse sequence the signal after three echoes spaced TE apart is similar to the signal that would be obtained in a single spin-echo sequence having a time to echo of $3TE$. That is, signals may be measured at several different TE values (in this example, TE, $2TE$, and $3TE$) to yield significantly more information about the sample than from a single sequence. This is only true, however, if TR is much longer than the total TE for n pulses (nTE , where n is the number of 180-degree rephasing pulses in the sequence). In practice, the storage space and computer time required to process multiple echo images have caused them to be used less often than spin-echo images that may be tailored to bring out one or more of the factors that determine image contrast.

Inversion Recovery

Inversion recovery (Margin Figure 24-5) takes its name from an initial 180-degree pulse that inverts spins (nutates them 180 degrees). At a time TI (inversion time) after the initial 180-degree pulse, a 90-degree pulse is applied about the same axis. After the 90-degree pulse the inversion recovery sequence is the same as spin echo. At a time T after the 90-degree pulse, a 180-degree rephasing pulse is given. As in spin echo, this 180-degree pulse produces a rephased signal at a time $TE = 2T$ after the 90-degree pulse. As in all pulse sequences, the time between one complete pulse sequence and the next is defined as TR. The 180-degree pulse used at the beginning of inversion recovery allows more time for differences in longitudinal relaxation of spins between pulse sequences than the smaller (e.g., 90-degree) pulses used in spin echo. Thus, differences in T_1 among tissues are, in principle, accentuated by the inversion recovery sequence.

SPATIAL ENCODING OF MAGNETIC RESONANCE IMAGING SIGNAL

In Chapter 23, techniques for receiving and manipulating signals from bulk magnetic moments were discussed. To form an image or to determine the origin of spectral information, the signals must be related to their points of origin within a sample. This mapping or identification of signals with regions of space is called *spatial encoding*.

Sensitive-Point Method

The signal from each volume element (“voxel”) in the patient could be received separately in sequence (i.e., one voxel at a time). One could then record the value of each signal in a separate storage location in the computer and reconstruct the image by displaying the values as a matrix of shades of gray. In practice, this method of data acquisition, called “sensitive-point” scanning,³ is too time-consuming, and patient motion would be unacceptable. It is described here to illustrate several fundamental points about data acquisition and to emphasize the need for more complex but efficient methods of acquiring data.

If the magnetic field is made to vary gradually across the patient, then according to the Larmor equation the frequency of precession and therefore the resonance frequency of the protons will also vary. Suppose that the variation or “gradient” of the magnetic field is in the craniocaudal direction (Margin Figure 24-6); that is, suppose the magnetic field is stronger at the patient’s head than at the feet. If an RF pulse of narrow bandwidth (a narrow range of frequencies) is sent into the patient while the gradient magnetic field is on, the pulse resonates with protons somewhere along the craniocaudal axis. Judicious selection of the gradient and the pulse frequency will result in resonance with protons in a selected transverse slice of the patient. The bandwidth or range of frequencies in the pulse will determine the thickness of the slice. Thus we have defined a plane within the patient from which we have placed the protons at some preselected orientation with respect to the plane of the receiver coils (e.g., with a 90-degree pulse). We could repeat the procedure in the other two orthogonal planes by using a dorsal–ventral magnetic field gradient to define a coronal plane and a left–right gradient to define a sagittal plane. The intersection of three planes is a point and yields a “sensitive point” for data acquisition.

When RF pulses of a finite bandwidth are used, the intersection of the three orthogonal planes forms a volume element, or voxel, of time. We may “communicate” with (receive MR signals from) this voxel because the location has a uniquely defined resonance frequency. This spatial encoding procedure of activating three orthogonal magnetic field gradients is then repeated for each voxel in a slice to obtain data to be displayed as a matrix of picture elements (“pixels”).

In practice, the sensitive-point technique has been implemented by using alternating gradient fields (Margin Figure 24-7). The effect of the alternating fields is to provide changing signals that can be recognized and discarded electronically at all locations except the “node” where the gradient is zero, leaving the spins to experience the static magnetic field of the imaging system. The location of the node can be defined by three orthogonal gradients and can be steered throughout the patient.

Example 24-1

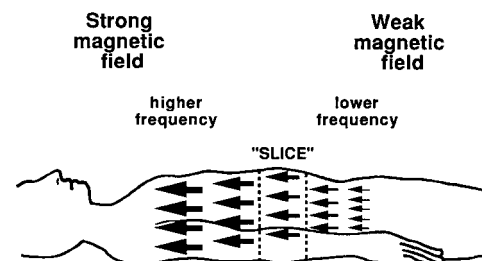
If it takes 300 msec to obtain a single voxel by using a sensitive-point technique, how much time is required to obtain five slices with a matrix size of 256×256 ?

$$\begin{aligned}\text{Imaging time} &= 256 \times 256 \times 5 \times 300 \times 10^{-3} \text{ sec} \\ &= 98,300 \text{ sec} \\ &= 27.3 \text{ hr}\end{aligned}$$

Two-Dimensional Fourier Transform

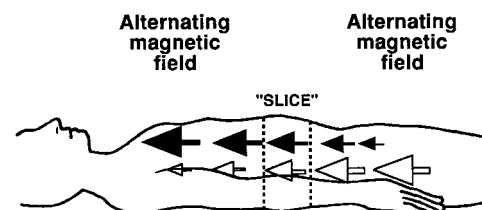
The sensitive-point technique makes use of “frequency encoding” to establish a unique resonance frequency for each voxel in the patient. The most obvious limitation of the sensitive-point technique is that excessive time is required to obtain enough information to construct an image. Fortunately, there are faster methods of spatial encoding. Signals may be encoded according to their phase. They may also be encoded by frequency a second time by using the Fourier transform. A number of alternate image acquisition schemes have been employed. These include sensitive-line,^{4,5}

In the sensitive point method, the “point” is actually a small volume element (voxel) of tissue with dimensions defined by the gradients and the frequency range of the RF pulse.



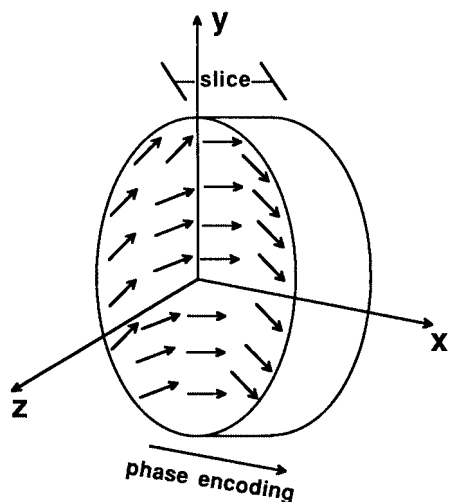
MARGIN FIGURE 24-6

A craniocaudal “gradient” or change in the magnetic field along the craniocaudal axis allows receiver coils to be “tuned in” to a slice.



MARGIN FIGURE 24-7

The sensitive-point technique begins with the application of alternating magnetic fields. For part of a cycle the magnetic field increases from right to left as shown here. The direction of increase then reverses. The region defined as the “slice” maintains the same magnetic field value.



MARGIN FIGURE 24-8

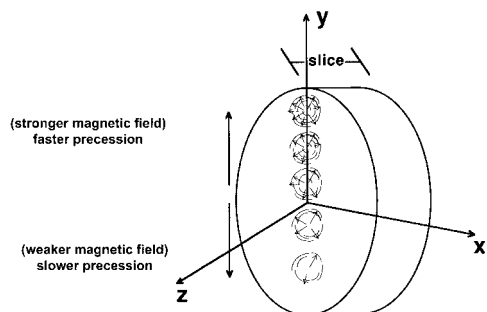
The effect of the phase-encoding gradient is to vary the phase of precession of the protons along the x direction. The phase of precession of protons along the y direction is the same at any location along the x -axis.

sequential-plane,⁶ and three-dimensional zeugmatography.⁷ At the present time, the most commonly used method of spatial encoding in commercial MRI systems is the two-dimensional Fourier transform (2DFT).⁸

The first step in the 2DFT method of signal acquisition is activation of a slice-select gradient while a narrow-bandwidth RF pulse is sent. As discussed above, the presence of the slice-select gradient means that protons in a narrow section are in resonance and therefore nutated to a specified angle by the RF pulse. Protons elsewhere in the patient are relatively unaffected. Within the plane, however, the angle of nutation depends upon the magnitude and duration of the RF pulse. The location of the plane along the direction of the gradient is determined by the center frequency of the RF pulse, and the thickness of the section or "slice" is determined by the bandwidth of the RF pulse.

As in any spatial encoding scheme, the 2DFT requires that information about the location of the signal be provided along three orthogonal axes. The first axis is "slice," as defined above, which is usually identified as the " z direction" and is oriented along the cranial-quadrant axis of the patient. The other two axes, x and y , are identified with or "encoded" by the frequency and phase of the RF signals returning from the tissue. The second step of 2DFT is to turn on a phase-encoding gradient in a direction perpendicular to the slice-encoding gradient. This gradient is turned off before any RF pulses are applied. In Margin Figure 24-8 the phase-encoding direction is along the x -axis, but this alignment is arbitrary.

It may seem that turning the phase-encoding gradient on and off between pulses is a useless exercise. When it is on, protons at the end of the gradient where the magnetic field is stronger precess faster (at a higher frequency) than do protons at the other end, with intermediate values of precession frequency in between. When the gradient is turned off, protons within the slice revert to a common precession frequency. Thus the phase-encoding gradient has a very significant effect upon the MR signal (Margin Figure 24-8). When the phase-encoding gradient is on, it divides the slice into columns with different precessional frequencies. When the gradient is turned off, the protons revert to the precession frequency called for by the static magnetic field provided by the main system magnet, and no difference in frequency exists from one column to the next. However, protons in the columns that earlier experienced a higher frequency of precession rotated farther during the time that the phase-encoding gradient was on, and protons in the columns that experienced a lower frequency of precession did not rotate as far during the same interval. Thus when the gradient is turned off, the magnetization is locked in with protons in different columns precessing at different phases of rotation (i.e., 12 o'clock, 1 o'clock, etc.), thus giving rise to the term *phase encoding*. That is, the difference between columns is encoded by the phase of rotation.



MARGIN FIGURE 24-9

The effect of the frequency-encoding gradient is to vary the rate or frequency of precession of protons along the y direction.

The third step in 2DFT is to apply a frequency-encoding gradient at the time that the RF signal is received from the patient. The frequency-encoding gradient acts according to the same principle as the slice-select gradient or the phase-encoding gradient. When it is on, protons at one end of the slice precess faster than do protons at the other end. The "fast" and "slow" ends define an axis or direction. In Margin Figure 24-9 the frequency-encoding gradient is identified with the y -axis. The effect of this gradient is to divide the slice into rows. Thus, while the RF receiver is turned on, protons in different rows precess at different frequencies, and the rows are encoded by the frequency of rotation. Because the frequency-encoding gradient is turned on only during signal reception or "readout," it is often referred to as a "read gradient."

The fourth step of 2DFT is the application of the Fourier transform (see Appendix I). This approach permits a complex data set to be acquired in a single set of measurements over a relatively short period of time. The data are analyzed later with a mathematical transformation or mapping between Fourier "transform pairs." In MRI, there are two transformations (two-dimensional Fourier transformation). In step 3 above, the MR signal from all columns is read while the frequency-encoding gradient is on. Because of this gradient, the MR signal is a complex signal made up of the contributions from protons in voxels having different frequencies of precession

(Margin Figure 24-10). The Fourier transform determines the contributions that were made by different pixels by transforming the MR signal (acquired as a function of time) to its mathematical representation as a function of frequency. Thus, the Fourier transform pair in step 3 consists of time and frequency. The data from different rows (different frequencies of precession) are stored separately in preparation for the second Fourier transform.

The fifth step is to repeat steps 1 to 4 a number of times and average the result. This “signal averaging” reduces the contribution of noise to the final data.

At this point we might think of the 2DFT as having resulted in the encoding of information (the MR signal) from one slice of voxels. The slice-select gradient has defined the slice. The phase-encoding gradient has defined columns, and the frequency-encoding gradient has defined rows. However, it is not quite so simple. A single application of the phase-encoding gradient does not provide enough information to fully differentiate the columns within a slice. The essence of the Fourier transform method is that a complex signal can be analyzed to determine the contribution of different elements to the signal. It is thus necessary to acquire data for the entire slice before both Fourier transforms may be applied.

The sixth step of 2DFT is to change the magnitude of the phase-encoding gradient and then repeat steps 2 to 5. The number of times that this step is carried out determines the number of columns into which the slice will be divided—that is, the number of voxels in the phase-encoding direction. Recall that along a row the voxels have different phases (see Margin Figure 24-8). The MR signal that is received after application of each phase-encoding gradient yields additional information about the contribution of different protons along the row. That is, the row can be divided more finely (i.e., into more voxels) by exposing it to more gradients.

In the seventh and final step of the 2DFT method the second Fourier transform operates on the complex signal obtained after all of the phase-encoding steps have been completed. This second transformation makes use of the transform pair of phase and position. That is, the contributions from the different phase combinations obtained from the separate phase-encoding steps (rows) are determined by Fourier transformation of the complex signal made up of all of the data.

The steps involved in the 2DFT method may be summarized as follows:

1. Apply the slice-select gradient while RF pulses are transmitted.
2. Apply the phase-encoding gradient.
3. Apply the frequency-encoding gradient while the MR signal is received.
4. Fourier-transform the received MR signal.
5. Repeat steps 1 to 4 and average.
6. Repeat steps 2 to 5 with different phase-encoding gradients.
7. Fourier-transform the data from steps 1 to 6.

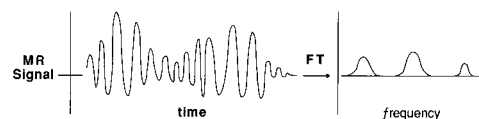
The 2DFT reduces scan time significantly compared with “sensitive-point” techniques. Mathematical analysis is required to determine the contribution of individual voxels within rows and columns to the total signal, but time spent in computation does not affect motion unsharpness in the final image. A diagram of the 2DFT method is shown in Margin Figure 24-11.

The acquisition time T_{aq} required to obtain an image by using 2DFT is determined by the number of signal averages N_a , the number of phase-encoding steps N_p , and the time that the imager has been programmed to wait to complete the pulse sequence that was selected, that is, TR (pulse repetition time). Thus

$$T_{aq} = N_a N_p TR \quad (24-1)$$

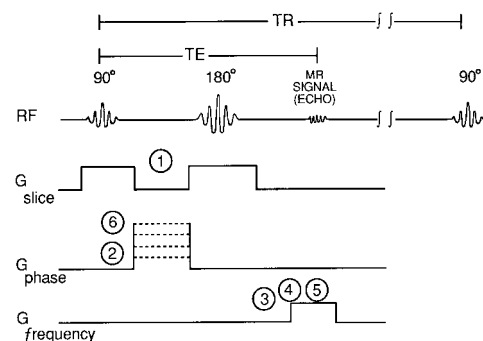
Example 24-2

Find the total time required to obtain an image if four signal averages and 256 phase-encoding steps are used in a pulse sequence that requires 1500 msec to complete.



MARGIN FIGURE 24-10

The Fourier transform (FT) of the MR signal acquired during the application of the frequency-encoding or “read” gradient. The MR signal as a function of time is transformed mathematically into relative magnitudes at different frequencies.



MARGIN FIGURE 24-11

The two-dimensional Fourier transform (2DFT) radio-frequency (RF) pulses and gradient magnetic fields (G) are shown on separate time lines illustrating the order in which events occur. The exact shapes of pulses and gradients are not represented. The steps involved in the entire 2DFT are as follows: (1) The slice select gradient (G_{slice}) is applied while RF pulses are sent; (2) a phase-encoding gradient (G_{phase}) is applied; (3) a frequency encoding gradient ($G_{\text{frequency}}$) is applied while an MR signal is received; (4) an MR signal is Fourier-transformed; (5) steps 1 to 4 are repeated, and the results are averaged; and (6) steps 2 to 5 are repeated with a different phase-encoding gradient (G_{phase}). A second Fourier transform is applied to the data set obtained in steps 1 to 6.

TABLE 24-1 MRI Motion Suppression Techniques

Technique	References
Breath holding, physical restraints	9,10
Respiratory gating	11,12
Data averaging	13–15
Respiratory ordering of phase encoding (ROPE)	16–18
Gradient-echo techniques	19–22
Gradient-moment nulling	23,24
Half-Fourier imaging	25
Snapshot	26

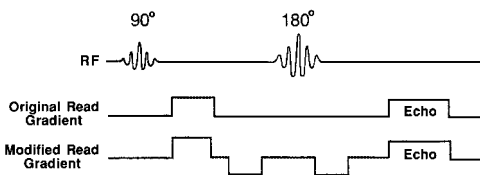
$$\begin{aligned} T_{\text{aq}} &= N_a N_p \text{TR} \\ &= 4 \times 256 \times 1500 \times 10^{-3} \text{ sec} \\ &= 1536 \text{ sec} \\ &= 25.6 \text{ min} \end{aligned}$$

MOTION SUPPRESSION TECHNIQUES

Various schemes have been proposed for suppressing motion artifacts in MRI. Table 24-1 lists several of these techniques. Methods of motion suppression may be divided into three groups: techniques that suppress motion in the patient, those that make novel use of scan data to remove the effects of motion, and the use of faster pulsing schemes.

Suppression of Motion of Patient

Methods such as patient restraint, sedation, acquisition during breath holding, and gated acquisition are obvious extensions of techniques that have had some success in other imaging applications. Ultimately, breath holding is practical for most patients only if scan times are reduced to the order of seconds. Respiratory gating with a mechanical device to time data acquisition during one or more phases of the respiratory cycle has the undesirable feature of an increase in overall examination time. Also, the image is a composite, and transient physiologic phenomena will not be seen.



MARGIN FIGURE 24-12
A simple modification was made to the spin-echo imaging technique to rephase magnetization moving at constant speed along the readout direction. Two identical lobes were added on either side of the 180-degree refocusing RF pulse. The two lobes must have similar polarity because the 180-degree pulse reverses the orientation of the transverse magnetization. Consequently, the area under the readout gradient is left unchanged. The size of the new lobes can be adjusted to achieve similar dephasing for stationary tissue and tissue moving at a constant speed. (From Runge, V. M., and Wood, M. L. *Magn. Reson. Imaging* 1988; 6:595–608. Used with permission.)

Removal of Effects of Motion

The order of phase-encoding steps may be selected according to the respiratory motion of the patient so that the end result is an image that appears to have been taken during a single respiration. There is some residual blurring with the respiratory-ordered phase-encoding technique (ROPE), but the reordering of data does not require intermittent scanning as in a gating technique.

Effects of motion that occur at a constant rate along the direction of a gradient may be removed by altering the gradient. In gradient-moment-nulling techniques, the shape of the gradient is changed over time (Margin Figure 24-12) so that symmetrical application of the gradient rephases spins that have moved a specified distance during a pulse sequence. In addition to allowing motion suppression in only the three gradient directions, gradient nulling places a limit upon minimum achievable TE because of the time required to reapply new gradients during a pulse repetition.

Fast Scanning Techniques

Two significant limitations on the time required for a pulse sequence are the repetition time (TR) required to allow recovery of longitudinal magnetization and the echo time (TE) mandated by use of a 180-degree rephasing pulse. Both of these limitations are addressed in gradient-echo imaging techniques.

As mentioned earlier, the purpose of the 180-degree pulse in a spin-echo sequence is to remove the effects of magnetic field inhomogeneities related to T_2^* . The gradient echo is another technique for dealing with this problem. In gradient-echo pulse sequences, while the phase-encoding gradient is on, the frequency-encoding gradient has a specified direction. For example, the magnetic field may be increased along the positive y-axis and decreased along the negative y-axis. After the phase-encoding gradient is switched off, the frequency-encoding gradient is reversed (e.g., decreased along the positive y-axis and increased along the negative y-axis). The effect of the reversal of gradient is the following: spins that experienced an increased field before reversal precess at a higher frequency and vice versa. Thus, a type of dephasing, albeit organized and directional, occurs. After gradient reversal, the faster spins become the slower spins and will be “caught” by the average spins. Thus, a type of rephasing occurs that results in a maximum signal that occurs at a predictable time after reversal of the frequency-encoding or read gradient (Margin Figure 24-13). In summary, the gradient echo does away with the 180-degree rephasing pulse by substituting a reversal of the magnitude of the frequency-encoding or read gradient.

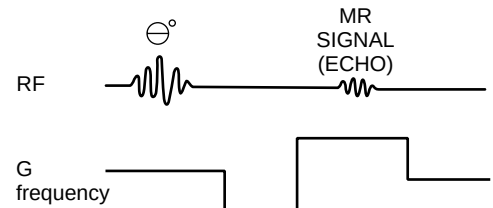
The gradient reversal does not accomplish all that the 180-degree rephasing pulse achieves because it does not remove the effects of magnetic field inhomogeneities. Therefore, the appearance of an image is influenced by T_2^* as well as T_2 . The effects of chemical shifts are also not removed by the gradient echo and can contribute to decreased signal intensity, especially at high field strengths. However, it performs the function of separating the echo, the maximum MR signal induced in the receiver coils, from the initial pulse. Because the initial part of the gradient, before the reversal, may be applied while the phase-encoding gradient is on, a reduction in TE is achieved when compared with the method of the 180-degree rephasing pulse.

The use of a gradient-echo technique allows a significantly shorter TR. An effect of shortening TR is that longitudinal magnetization does not have time to recover completely between pulse sequences. When complete recovery does occur, a 90-degree pulse provides maximum transverse magnetization and therefore a maximum induced signal in the receiver coils. During a pulse sequence in which TR is too short to allow complete recovery of longitudinal magnetization, however, a 90-degree pulse nutates the net magnetization past the transverse plane and provides a reduced signal. Thus, for techniques such as gradient echo where TR is shortened, flip angles of less than 90 degrees result in an improved signal-to-noise ratio.^{27,28}

Examples of gradient-echo techniques include GRASS (gradient-recalled acquisition in the steady state),²⁹ FLASH (fast low-angle shot imaging),³⁰ and FISP (fast imaging with steady-state precession).³¹

Half-Fourier Technique

There is symmetry in the data produced by repeated application of the phase-encoding gradient (the phase-encoding “steps” of signal acquisition).²⁵ In theory, it is necessary to acquire only half of the steps, say those produced with a negative gradient, with the other half, the positive gradient steps, reproduced mathematically from those acquired. This half-Fourier technique saves imaging time, with the only drawback in theory being an increase in computer reconstruction time. In practice, however, use of the second half of the phase-encoding steps is a method of signal averaging (multiple measurement of the same data). Because the half-Fourier technique uses fewer data, the signal-to-noise ratio may be reduced.



MARGIN FIGURE 24-13

Gradient-echo imaging technique. The radio-frequency pulse (RF) and frequency-encoding magnetic field gradient ($G_{\text{frequency}}$) are shown. Other gradients are similar to those used for spin-echo 2DFT image acquisition. The frequency-encoding gradient is reversed to bring about a rephasing of spins. This technique eliminates the 180-degree pulse used in spin echo. The RF pulse used in gradient-echo techniques is usually less than 90 degrees.

Time-Varying Gradient

In standard imaging techniques the phase-encoding gradient is stepped through 128 or 256 discrete values, with the entire pulse sequence repeated for each value. In snapshot imaging, all the phase-encoded values are collected during a single RF pulsing sequence. Spatial encoding is accomplished by rapid oscillation of the frequency-encoding gradient interspersed with pulses of the phase-encoding gradient that occur as the x gradient returns to zero between oscillations. Rapid data sampling during each excursion of the frequency-encoding or read gradient results in the final signal acquisition. To date, snapshot techniques have yielded 64×128 images in 1/50th of a second.²⁶

■ CONTRAST AGENTS

There are three basic classifications of materials with regard to magnetic properties: paramagnetic, diamagnetic, and ferromagnetic.

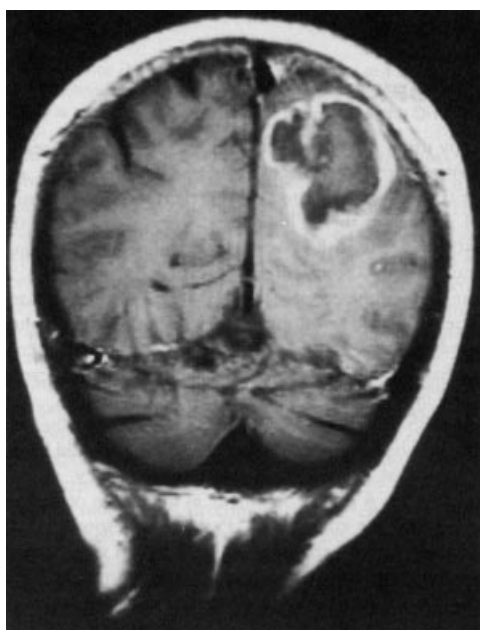
Electrons have spin and magnetic moments just as do nucleons. In fact, the magnetic moment of the electron is more than 100 times greater than the magnetic moments of nucleons. Their effect is less noticeable, however, because the magnetic moments of electrons in orbitals or shells surrounding the nucleus tend to “pair up”—for example, one electron having spin up, the other having spin down. In diamagnetic substances, virtually all electrons are paired, so interactions with external magnetic fields are minimized. In fact, diamagnetic substances actually repel an external magnetic field, albeit weakly. The repulsion is due to an induced magnetic field in the electron orbital. Such an induced field opposes the field that induced it (Lenz’s law).³² Approximately 99% of biologic tissues are diamagnetic. Because of their high natural abundance in body tissue, there is little current interest in the development of diamagnetic substances as contrast agents, although diamagnetic clay suspensions have been used for contrast in the gastrointestinal (GI) tract³³ by eliminating signal from the bowel contents.

Paramagnetic contrast agents contain unpaired electrons. They develop their own magnetic fields by tumbling in solution at some fraction of the Larmor frequency. The tumbling creates alternating magnetic fields that affect precessing nuclei. An interaction between the unpaired electrons and the precessing nuclei increases the rate of relaxation of neighboring protons, and the presence of variations in the magnetic field introduced by the paramagnetic agent increases dephasing. Thus, both T1 and T2 are shortened by the presence of a paramagnetic agent.

Only a small fraction, less than 0.5%, of the body’s tissues are naturally paramagnetic. Paramagnetic agents may be added to enhance relaxation and provide a mechanism for contrast enhancement. Thus, paramagnetic contrast agents are not imaged directly in MRI. Instead, the presence of the agent influences the appearance of the image. One of the most widely used paramagnetic agents is gadolinium with diethylenetriamine pentaacetic acid as a chelating agent (Gd-DTPA).³⁴ An example of contrast enhancement by Gd-DTPA is shown in Margin Figure 24-14.

Ferromagnetic materials develop magnetic polarization (i.e., become magnetic) in the presence of an applied magnetic field. Ferromagnetism is a phenomenon in which hundreds to thousands of atoms having unpaired electrons group together into “domains.” If many domains are aligned, then the material is said to be magnetized. If the alignment persists in the absence of an applied magnetic field, then the material is a permanent magnet. Because a large grouping of atoms is required for ferromagnetism, ferromagnetic contrast agents do not exist as solutions of metal ions. They are given in the form of compounds such as iron oxides (Fe_2O_3 , Fe_3O_4). These agents selectively decrease T2 and cause a decrease in MR signal in conventional pulse sequences. Thus ferromagnetic MR contrast agents provide “negative” contrast.

In addition to diamagnetic, paramagnetic, and ferromagnetic mechanisms of contrast agent action in MR, other mechanisms are possible. Perfluorochemicals, substances in which all or part of the hydrogen has been replaced by fluorine, elicit no MR signal in conventional proton imaging because they contain no hydrogen. These



MARGIN FIGURE 24-14
An example of contrast enhancement by Gd-DTPA.

substances have been used to provide negative contrast.³⁵ Similarly, the introduction of gas-evolving substances and of air itself into the body has been investigated.³⁶

Functional MRI (fMRI)

Oxygenated blood, like most body tissues, is diamagnetic despite the presence of iron-containing hemoglobin. In oxygenated hemoglobin (oxyhemoglobin), the outer-shell electrons of the iron atom are transferred to oxygen molecules. In deoxygenated blood (deoxyhemoglobin), four of the six outer electrons of the heme ion are unpaired, placing the iron in a ferrous (Fe^{2+}) state that is paramagnetic. In the absence of neuronal activity, the presence of paramagnetic deoxyhemoglobin in red blood cells causes a susceptibility gradient (change in the effective magnetic field over a small region of space) that has the effect of shortening T2 and T2* in the surrounding tissue and blood plasma. These regions of magnetic field inhomogeneity produce measurable effects over distances of two or more times the radius of the vessel.³⁷

It is known that neuronal activity causes a localized increase in blood flow, cerebral blood volume, and oxygen delivery.³⁸ However, oxygen extraction increases by a lesser amount, if at all.^{39,40} The increased blood flow “drives out” the deoxyhemoglobin so that there is a net *decrease* in the concentration of deoxyhemoglobin in capillaries and venous circulation. The susceptibility gradient that exists between oxygenated blood and surrounding tissues in the *absence* of neuronal activity is reduced. Therefore, in the area of neuronal activity, less intravoxel dephasing occurs. The effect of this reduction in dephasing is a net increase in signal from tissue in the region of neuronal activation in T2- and T2*-weighted images. Such signal intensity changes are observable within seconds after the onset of stimulation of neuronal activity.⁴¹ The time course of the rise of signal intensity is consistent with cerebrovascular transit times measured with radiotracer studies,⁴² supporting the belief that the increase in signal intensity is related to the transition of relatively oxygenated blood through the capillary bed and into venules. Thus, the presence of a naturally occurring contrast agent (oxyhemoglobin) that responds to the spatial and temporal pattern of neuronal activation in the brain allows construction of a functional map of neuronal activity (see Margin Figure 24-15). The spatial resolution of this functional map is limited by the volume of the region of decreased oxyhemoglobin concentration. Temporal resolution is limited by the consistency of the several second lag time between onset of neuronal stimulation and the rise in MR signal.

Functional MRI (fMRI) has revolutionized the study of brain function. It has upgraded brain research from a field in which the approximate anatomical correlates of function were known through studies of individuals with neurological abnormalities or injuries, to a field in which it is possible to determine the specific locus of many neural activities in specific individuals.^{44,45} This technique is replacing presurgical mapping^{46–53} to identify the location of functional areas before tumor resection, and it is valuable in patient evaluation following stroke or other neurovascular events.^{54–57} It has already led to major advances in understanding the linkage between structure and function in the cognitive sciences,^{58–63} particularly in the study of learning and development.^{64–66} Margin Figure 24-16 shows a functional MR study in which the areas associated with motor, memory, and visual processing are examined during successive tasks in which the subject moves an index finger in response to movement of a cursor on a screen, or simply watches the movement of the cursor. Included in the illustration is the image formed by subtraction of the two acquired images to isolate the role of vision in a simple task.

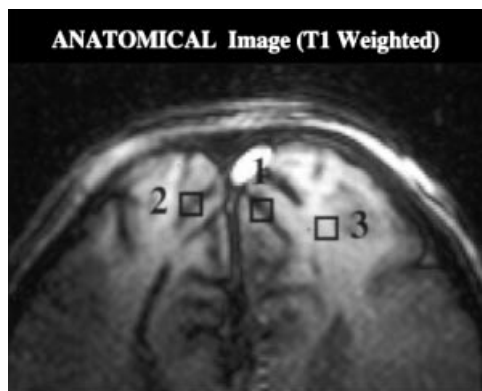
Magnetic Susceptibility is a property of matter that modifies an externally applied magnetic field so that an induced magnetic field appears in and around the matter. A related concept using a slightly different scale is **magnetic permeability**. A high value of susceptibility (or permeability) indicates that the material will “act like a magnet” when placed in a magnetic field. Such material is classified as **ferromagnetic**. Iron is an example of a ferromagnetic material. Some matter has negative magnetic susceptibility—it actually tries to repel a magnetic field (although the repulsive force is generally too weak to notice). Such materials are classified as **diamagnetic**. In fact, most biological materials, water, most organic molecules, and so on, are diamagnetic.

In MRI, variation in the actual magnetic field that is induced in a region of the patient is produced by variation in the susceptibility of adjacent structures. These **susceptibility gradients** between adjacent structures such as brain/sphenoid sinuses and CSF/vertebral bodies can cause artifacts and signal losses. Susceptibility gradients can also enhance the appearance of structures by changing local relaxation times in comparison with the surrounding materials where materials such as gadolinium are introduced as contrast agents.

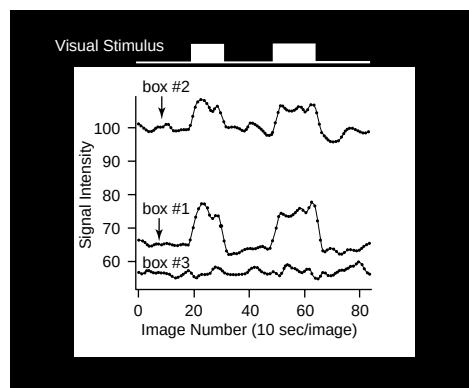
Seiji Ogawa of Bell Labs was the first to propose the technique of functional MRI, also known as blood-oxygen-level-dependent (BOLD) imaging. He published his work on animal studies and suggested the application of this technique for imaging of brain function in humans in 1990. He collaborated with the Center for Magnetic Resonance Research at the University of Minnesota⁴³ to produce some of the earliest functional MR images made with human subjects.

TISSUE CONTRAST IN MAGNETIC RESONANCE IMAGING

Spatial encoding of MR signals is pointless if there are no differences in magnitude among the signals. Fortunately, there are substantial differences among human tissues in relaxation times (T1, T2) and proton density $N(\text{H})$. To illustrate concepts that



(a)



(b)

MARGIN FIGURE 24-15

An early indication of the utility of functional magnetic resonance imaging (fMRI) in mapping areas of brain function. A visual stimulus is shown to a research subject. In the T1-weighted MR image, areas 1 and 2 are known to be involved in processing visual stimuli and area 3 is not. A plot of fMRI signal intensity as a function of time shows that the signal increased during visual stimulus in the two areas associated with processing of visual stimuli, but did not increase in area 3.

are central to measurement of contrast in MRI, a rudimentary “patient” or sample containing only two materials is considered below, each having characteristic values of T_1 , T_2 , and $N(H)$. With fixed characteristics of the imaging system (e.g., field strength and receiver coil geometry) the imaging experiment consists of choosing operator-selectable parameters (TR , TE , TI , flip angle, etc.) that maximize the contrast or difference in signal between the two materials.

Spin Echo

The strength intensity S_{SE} of the MR signal during a spin-echo sequence is given by

$$S_{SE} = N(H) \left[1 - 2e^{-(TR - TE/2)/T_1} + e^{-TR/T_1} \right] e^{-TE/T_2} \quad (24-2)$$

“Scaling factor”
“Exponential growth”
“Exponential decay”

where $N(H)$ is the number of spins, T_1 is the longitudinal relaxation time constant, T_2 is the transverse relaxation time constant, TE is the time to echo (time between the 90-degree pulse and rephased signal or echo, which is twice the time between the 90- and 180-degree pulses), and TR is the pulse sequence repetition time.

The influence of T_1 in this equation depends on the magnitude of TR . T_1 and TR appear together in an “exponential growth” term. Spin density, $N(H)$, appears as a scaling factor or multiplier and has a linear effect upon signal strength. That is, if the spin density is doubled, the signal strength is doubled.

The time between pulse repetitions, TR , allows the bulk magnetization of a material to return to alignment with the axis of the static magnetic field (the z -axis). If alignment is complete, then the next 90-degree pulse nutates the bulk magnetization into the xy plane to yield a maximum signal. Margin Figure 24-17, A, where signal strength is plotted as a function of TR from Eq. (24-2) (assuming constant TE), shows the exponential growth of signal with TR . The magnitude of the maximum signal (when TR is infinitely long) is determined by the total number of spins contributing to the signal, that is, spin density $N(H)$. Also shown in Margin Figure 24-17A is the effect of T_1 . If the only difference between two materials is T_1 , then the material having the shorter T_1 would produce a stronger signal at intermediate values of TR .

The time between the 90-degree pulse and the echo, TE , determines the degree to which dephasing of the spins, T_2 relaxation, is allowed to reduce the transverse or “ xy ” component of the bulk magnetization. If the dephasing is allowed to progress to the point where the phase of spins is random, the signal becomes zero. Margin Figure 24-17B, where signal strength is plotted as a function of TE from Eq. (24-2) (assuming constant TR), shows the exponential decay of signal with TE . The magnitude of the maximum signal (when TE is zero) is determined by spin density, $N(H)$. Also shown in Margin Figure 24-17B is the effect of T_2 . Note that if the only difference between two materials were T_2 , then the material having the longer T_2 would produce a stronger signal at intermediate values of TE .

If the goal of imaging were to maximize the signal from a particular tissue, longer TR values and shorter TE values would always be preferred for spin-echo imaging. To obtain an image, however, one must obtain signals from a variety of tissues and maximize the contrast, or relative signal difference, among tissues of interest. Margin Figure 24-17A shows that the difference between signals for two materials having different values of T_1 (the vertical distance between the two curves at any TR) is maximum for TR near the average T_1 of the two tissues. Similarly, Margin Figure 24-17B shows that the difference between signals for two materials having different values of T_2 is maximum for TE near the average T_2 of the two tissues.

TABLE 24-2 Rules of Thumb for Contrast in Spin-Echo Imaging

TR	TE	Weighting	Stronger Signal Is Associated with
Short (~avg. T1)	Short	T1	Shorter T1
Long (>4 T1)	Short	N(H)	Greater N(H)
Long	Moderate (~ avg. T2)	T2	Longer T2

Summary of Contrast in Spin-Echo Pulse Sequences

It is possible to generalize about the role of TR and TE in bringing out the contrast between tissues. Table 24-2 gives some rules of thumb for MR contrast. The terms “long, short, and moderate” are of course relative and must be defined in terms of the T1 and T2 of the materials of interest. Note that because the convention in imaging is to map higher signal values into brighter pixel values, the brighter pixels are correlated with shorter T1, longer T2, or higher spin density in spin-echo images.

The reader is cautioned that there are virtually no universally applicable rules for contrast optimization in MR. It is illustrative to consider the number of distinct cases that are possible when two materials that differ in one or more MR parameters [T1, T2, or N(H)], and T2 can be either greater than, less than, or the same. Margin Figure 24-18 shows the range of cases generated by selection of different values of TR and TE in a spin-echo sequence. By selecting different values of TR and TE, the operator can reverse contrast or eliminate it altogether.

Inversion Recovery

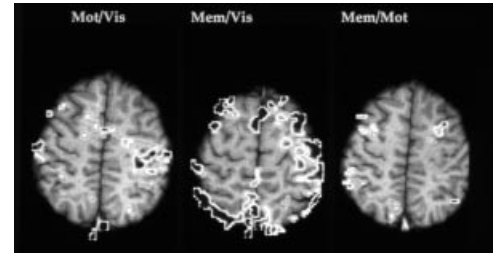
The strength intensity S_{IR} of the MR signal during an inversion recovery pulse sequence is given by

$$S_{IR} = \underbrace{N(H)}_{\text{“Scaling factor”}} \underbrace{\left[1 - 2e^{-TR/T1} + 2e^{-(TR - TE/2)/T1} - e^{-TR/T1} \right]}_{\text{“Exponential growth”}} \underbrace{e^{-TE/T2}}_{\text{“Exponential decay”}} \quad (24-3)$$

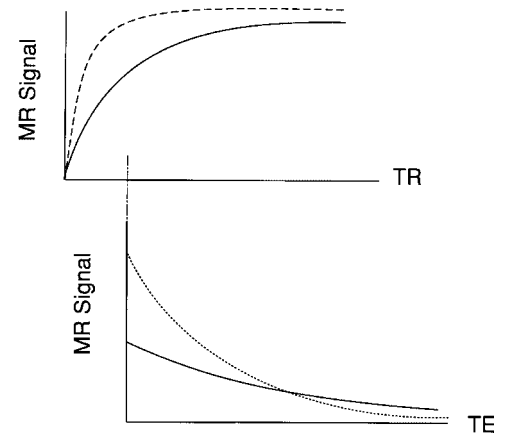
where $N(H)$ is the number of spins, T1 is the longitudinal relaxation time constant, T2 is the transverse relaxation time constant, TE is the time to echo (time between the 90-degree pulse and the rephased signal or echo, which is twice the time between the 90- and 180-degree pulses), TR is the pulse sequence repetition time, and TI is the inversion time (time between the initial 180-degree pulse and the next 90-degree pulse).

The equation for inversion recovery is similar to that for spin echo. The difference is a more complicated exponential growth term for the inversion recovery equation. As with the spin-echo pulse sequence, TE influences the degree of T2 weighting. In inversion recovery, however, the degree of T1 weighting is influenced by both TR and TI. If TI is sufficiently short in comparison with T1 and TR, a negative value of the signal is obtained. Margin Figure 24-19 shows signal and contrast as a function of TI for signal-processing schemes that preserve the sign, positive or negative, of the signal (“signed reconstruction”) and for those that do not (“absolute value” or “magnitude” reconstruction).

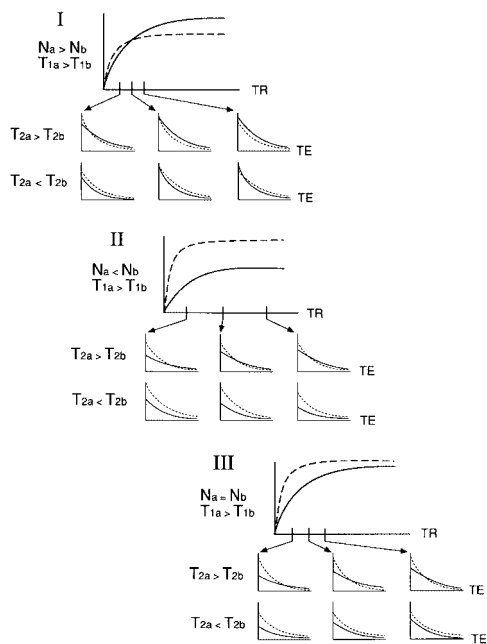
The same general rules for contrast optimization apply for inversion recovery as for spin echo, with the added factor of TI. Short TI enhances T1 by taking advantage of contrast reversal in magnitude reconstruction and is used in pulse sequences such as short TI inversion recovery (STIR).⁶⁷

**MARGIN FIGURE 24-16**

A series of functional magnetic resonance (fMRI) images taken during a study of learning ability and its relation to short-term memory activity. In the first image, the subject is given a task that involves motor skills and the visual system (to press a key in response to the appearance of an object on a view screen). In the second, the subject is told to remember a sequence as different objects are shown on a screen. The third image is obtained by subtracting the previous two under the hypothesis that it represents what was common to both tasks. Although not definitive in itself, this study is one of a number of novel approaches to new areas of study in understanding brain function that has been made possible with fMRI.

**MARGIN FIGURE 24-17**

Simulated MR signals from solution of the equation for spin echo [Eq. (24-2)] as TR and TE are varied. **A:** MR signal from two tissues with different T1 relaxation times (solid line, 600 msec; dashed line, 200 msec) as a function of pulse repetition time, TR, in spin-echo pulse sequence. TR varies from 0 to 2800 ms. Contrast (difference in MR signal, i.e., vertical distance between the curves) varies with TR. The two tissues shown here have the same spin density and therefore reach the same maximum signal value at long TR. **B:** MR signal from two tissues with different T2 relaxation times (solid line, 75 msec; dashed line, 40 msec) as a function of time to echo (TE) in a spin-echo pulse sequence. Contrast varies with TE. The two tissues shown here have the T1 and spin-density characteristics of the tissues in part A with a TR of 600 msec.



MARGIN FIGURE 24-18

Simulated MR signals showing contrast (vertical distance between curves) for spin-echo pulse sequences for two tissues having different relative values of spin density, N , longitudinal relaxation, T_1 , and transverse relaxation, T_2 . Tissue a (solid line) has a longer T_1 than tissue b (dashed line). The 21 graphs in this figure show all possible cases for N and T_2 . Each set is plotted first as a function of TR and then as a function of TE for short, medium, and long TR . Set I begins with $N_a > N_b$, Set II with $N_a < N_b$, and Set III with $N_a = N_b$. Contrast varies as a function of repetition time, TR , and time to echo, TE , for each case. Note that a pulse sequence may be selected for any case such that the resulting MR signals provide positive contrast (signal a > signal b), negative contrast (signal a < signal b), or no contrast (signal a = signal b). Signal values were calculated from the equation for spin echo [Eq. (24-2)] by Manhot Lau, Ph.D., University of Minnesota Medical School.

The main determinants of contrast in MRI are:

- Spin density
- T_1
- T_2
- Flow
- Magnetic susceptibility

Gradient Echo

When gradient reversals replace 180-degree pulses in sequences where flip angles of less than 90 degrees are used, contrast weighting depends primarily upon flip angle and TE . In general, for flip angles greater than 45 degrees with short TE (less than the average T_2), a T_1 weighting is achieved. For flip angles of less than 20 degrees, weighting is shifted toward T_2 and spin density.^{19–22}

MR ANGIOGRAPHY

Selective visualization of blood vessels with MRI may be accomplished with or without contrast agents. In single slice imaging, protons that flow into the slice may not have received the RF pulses that prepare the in-slice protons for the normal imaging sequence. This phenomenon may be exploited to image flowing blood. For example, protons in blood components that were outside of the slice during the application of an initial 90-degree pulse in a spin-echo sequence have not had their bulk magnetic moments “saturated”—that is, flipped to the angle that will produce a maximum signal for that number of spins. In time-of-flight MR angiography (TOF-MRA), the spins entering the slice represent blood components that are unsaturated, and these spins are in alignment with the static magnetic field rather than at 90 degrees to it. T_1 -weighted images pick up the blood as a stronger signal than its surroundings so that the vessels appear white.

Because blood flow is rarely exactly perpendicular to the plane of slices of interest, various 2D and 3D TOF techniques are used where several slices (or an entire slab) of tissue are imaged together to provide a higher signal-to-noise ratio. During these techniques, a series of 90-degree pulses is used to saturate blood outside of regions of interest. Lower flip angles are also used to avoid oversaturating the signal prior to the end of the sequence.

Irrespective of how the information is obtained, once a volume of tissue has been imaged in MRA, single slice views or volume-rendered images may be obtained by using an algorithm that picks the strongest signal acquired along a particular projection angle. This technique is known as maximum-intensity projection, or MIP (see Margin Figure 24-20).

Contrast agents such as gadolinium may be used to further enhance the signal from the blood vessels. By shortening the T_1 of the surrounding blood, the presence of gadolinium helps maintain the spins in their unsaturated state. However, gadolinium leaks quickly into surrounding tissues (except in the brain), so fast-scan techniques must be used. Many of the current generation fast-scan techniques push the envelope of acceptable levels of radiofrequency power levels (see Chapter 25). Thus, there is a trade-off between an increased signal-to-noise ratio and the desire to use more phase-encoding steps or number of slices.

SPECTROSCOPY

Spectroscopic applications of nuclear magnetic resonance (NMR) may be performed with some imaging systems. If performed in an MRI device, the technique is usually referred to as magnetic resonance spectroscopy (MRS). This technique has been used in analytical chemistry since the 1940s. MRS measures the differences in resonance frequencies among nuclei that occupy different positions in molecules. For example, protons in the CH_3 group of ethyl alcohol (Margin Figure 24-21) resonate at a slightly different frequency than do protons in the CH_2 group or protons in OH. The difference is only about $4 \times 10^{-4}\%$ (4 ppm) when the field strength is 1 tesla, much smaller than the differences in resonance frequency that are exploited for spatial encoding in MRI. Nevertheless, this difference can be measured.

In a typical MRS experiment, a radio-frequency wave is applied to the sample, and the signal is received. The naïve approach to spectroscopy would be to measure the amplitude of the signal, make a slight change in frequency, measure the amplitude again, and so on. A plot of amplitude as a function of frequency, the MR spectrum, could then be constructed. Peaks in the spectrum, frequencies at which a large amount of RF energy was absorbed and then returned by nuclei, correspond to the resonance frequencies of different nuclei. The resolution of the spectrum would be determined by the ability of the system to record the precise frequencies of radio waves and by the amount of time that is available for the experiment.

The time required for the spectroscopic approach described above is prohibitive. Instead of sending a radio wave with a single frequency, a radio wave containing a broad range of frequencies (a broad-bandwidth pulse) is sent. The return or re-emitted signal also contains a range of frequencies. The return signal may be Fourier-transformed to determine how much (what amplitude) of each frequency is contained within it. Thus, a single broadband pulse and receive sequence can replace the laborious set of measurements used in the more naïve approach. Some of the time that was “saved” is usually reinvested to improve resolution. The results of interrogation of the sample with hundreds of pulses are averaged to reduce the effects of noise.

The NMR spectrum for a molecule or compound is a unique, reproducible “signature” of great utility in determining the presence of unknown compounds in analytical chemistry. The origin of the unique pattern of chemical shifts for a molecule is the distortion of the magnetic field caused by the distribution of electrons that corresponds to each chemical bond. The effective field felt by each nucleus in the molecule is determined by the chemical bonds that surround it. Therefore, their resonance frequencies are shifted slightly. The individual protons within a chemical group experience slight differences in chemical shift as well because a hydrogen nucleus in the center of the group is influenced more strongly by the field of other hydrogen nuclei than is one at the periphery of the group. Therefore, a resonance peak for a chemical group is often split slightly into a number of equally spaced peaks, referred to collectively as a multiplet.

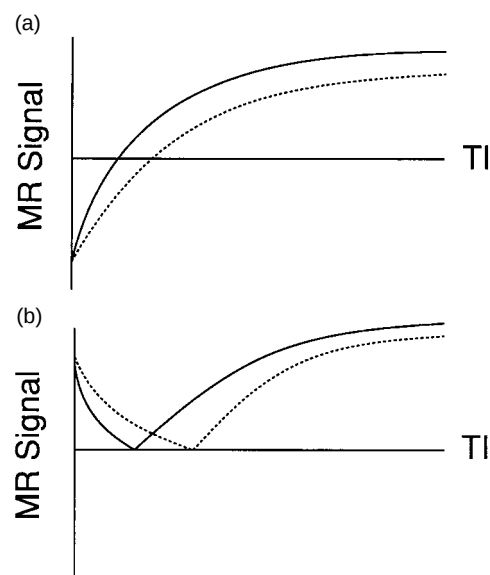
The chemical shifts discussed in this section so far are on the order of 10^{-7} tesla for a nominal magnetic field strength of 1 tesla. For spatial localization in imaging, gradient magnetic fields that produce differences of 10^{-4} tesla/cm are used. Therefore, the shift in resonance frequency produced by the magnetic field gradient would be 1000 times greater than the chemical shift and would completely overshadow it. One of the more significant differences between MRI and MRS is that during MRS no magnetic field gradients are applied. Furthermore, the homogeneity of the magnetic field over the MRS sample must be better than 1 part in 10 million.

According to the Larmor equation, resonance frequency depends upon the field strength applied to the sample. For this reason, chemical shifts are usually expressed as fractions, specifically as parts per million. That is, the resonance values are not expressed in terms of the absolute value of the resonance peaks f_P , but are described relative to some reference value of resonance frequency f_R and normalized to the nominal frequency of the system f_N according to the formula

$$\text{Chemical shift (ppm)} = \frac{f_R - f_P}{f_N} (10^6) \quad (24-4)$$

Example 24-3

An MRS system assumes a nominal frequency of exactly 300 MHz, the approximate resonance frequency for protons in a 7-tesla magnetic field. The peak against which other resonances are compared occurs at 300.004 MHz. Find the chemical shift in parts per million for a peak that occurs at a resonance frequency of 299.998 MHz.



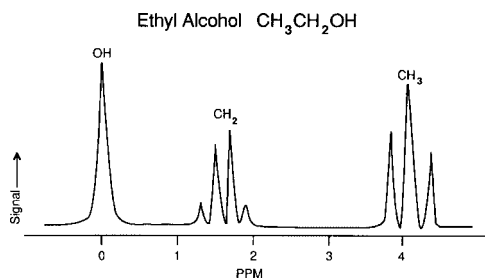
MARGIN FIGURE 24-19

Simulated MR signals from solution of the equation for inversion recovery [Eq. (24-3)] as inversion time, TI, is varied for two tissues. The longitudinal relaxation time T1 for the tissue represented by the solid line is shorter than T1 for the tissue represented by the dashed line. In **A** the sign of the signal (positive or negative) is preserved, while in **B** the absolute value (always positive) is retained. Spin density and T2 are equal.



MARGIN FIGURE 24-20

Maximum Intensity Projection (MIP) and surface rendering of an aortic aneurysm from magnetic resonance image (MRI) data. (Courtesy of Chun Yuan, Ph.D., Department of Radiology, University of Washington.)

**MARGIN FIGURE 24-21**

Nuclear magnetic resonance (NMR) spectrum of ethyl alcohol at 1 tesla. The chemical shift is expressed as a fraction (parts per million) of the reference frequency. In this case the reference frequency is the resonance frequency of the proton in the OH group.

$$\begin{aligned}
 \text{Chemical shift (ppm)} &= \frac{f_R - f_P}{f_N}(10^6) \\
 &= \frac{300.004 \text{ MHz} - 299.998 \text{ MHz}}{300.000 \text{ MHz}}(10^6) \\
 &= 0.00002 \times 10^6 \text{ ppm} \\
 &= 20 \text{ ppm}
 \end{aligned} \tag{24-4}$$

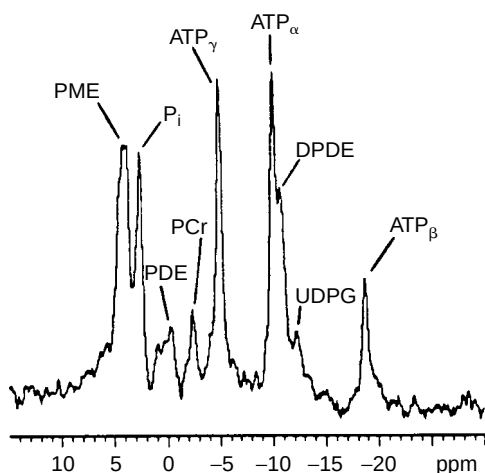
Other features of a spectrum, in addition to chemical shift, carry information about the sample. The area under a peak is proportional to the number of nuclei that contribute to the signal. The “line width” or full width at half-maximum (FWHM) amplitude is a measure of both the relaxation time in the sample and the degree of magnetic field inhomogeneity across the sample. Both factors tend to increase the range of frequencies gathered from equivalent nuclei during an experiment. A narrower line width is generally desirable, however, so that closely spaced peaks may be more readily resolved.

Nuclei Used for *In Vivo* Spectroscopy

Various nuclei may be examined with MRS. Table 23-1 shows the relative sensitivity for the same number of nuclei of interest.

³¹P

High-energy phosphate reactions within the cell may be traced with the aid of ³¹P spectroscopy. The phosphocreatine molecule is a storage site for high-energy phosphate groups used in metabolism and is often taken as the reference for chemical shift because its position does not change with pH.⁶⁸ The shift of inorganic phosphate with respect to phosphocreatine is then used as an indication of tissue pH.⁶⁸ The relative heights of the inorganic and phosphocreatine peaks can indicate changes in the amount of substances necessary for cellular metabolism (Margin Figure 24-22). It is also thought that extracellular and intracellular components of the same metabolites differ in their chemical shifts. Therefore, compartmentalization studies, which are difficult to perform analytically, may be conducted spectroscopically.⁶⁹ Enzyme kinematics may be studied through selective excitation of individual peaks and measurement of the intensity and relaxation time of neighboring peaks.⁷⁰

**MARGIN FIGURE 24-22**

³¹P spectrum of normal liver obtained through the anterior abdominal wall of an intact rat at 7 tesla using multivoxel localization with subsequent liver voxel addition. The resonances for phosphomonoesters (PME) and inorganic phosphate (P_i) are prominent relative to the phosphocreatine (PCr) resonance, which indicates that a small amount of muscle tissue was present within the volume assayed. Also noted are well-resolved peaks for diphosphodiester (DPDE), including uridine diphosphoglucose (UDPG). Phosphodiester (PDE) resonances are not prominent at this field strength. (Courtesy of Sally Weisdorf, M.D., University of Minnesota Medical School.)

¹H

Hydrogen or proton MR spectra may be used to observe metabolites such as lactate and glucose. An increase in the presence of lactate implies anaerobic conditions that may indicate ischemia or hypoxia. The peak from protons in water tends to obscure the peak from metabolites because it represents a 10,000-fold greater concentration of protons. However, suppression techniques are available to reduce the water peak through saturation (elimination of the net magnetization) of the water component or through selective excitation of the metabolites. The presence of the lipid peak also disrupts the observation of metabolites, which renders studies difficult unless performed in the brain where there is little detectable lipid.

²³Na, ³⁹K

Sodium and potassium are the major extracellular and intracellular cations, respectively. Studies of active transport mechanisms in nerve cells and other tissues are being pursued.^{71–73}

¹³C

The 1% isotopic abundance of ¹³C renders observation somewhat difficult. However, spectra have been obtained that show promise as probes of brain proteins⁷⁴ and for the detection of glycogen.⁷⁵ Intermediary metabolism of compounds such as pyruvate, ethanol, or glycerol enriched with ¹³C^{76,77} has also been investigated.

¹⁹F

The virtual lack of natural abundance of ¹⁹F in the body precludes spectroscopy of that material as a component of tissue. However, this abundance does provide an opportunity for use of ¹⁹F as a contrast agent. Fluorinated compounds have been used as blood substitutes.⁷⁸ Anesthetics containing fluorine have also been investigated and show promise as lipid-soluble compounds that may be suitable probes of membrane permeability.⁷⁹ One interesting feature of ¹⁹F is the similarity of its gyromagnetic ratio to that of hydrogen (see Table 24-1). Thus a suitable concentration of ¹⁹F may be useful for imaging in current scanners whose electronics are tuned for the reception of hydrogen.

■ CHEMICAL SHIFT IMAGING

In principle, the information obtained from the spectra of volume elements, or voxels, within a sample may be used to build up an image of the sample. In standard MRI, the spin density $N(H)$ and the relaxation parameters T1 and T2 determine the brightness of picture elements. These picture elements, or pixels, correspond to volume elements in the sample or patient. In chemical shift imaging (CSI), spectral information may be used directly to construct an image of a patient. A number of factors cause chemical shift imaging to be more difficult than proton imaging. In particular, the effects of magnetic field inhomogeneities tend to overshadow the small variations in resonance produced by the chemical shift.

Example 24-4

The separation of the main spectral peak due to water and the peak due to lipids (fat) is approximately 3 ppm. If the static magnetic field strength of an MRI unit is 1.5 tesla, what is the frequency shift in hydrogen between water and lipid?

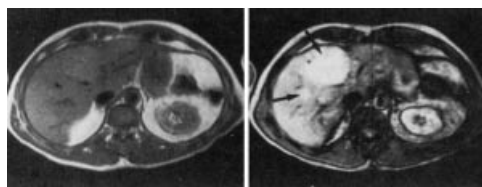
$$\text{Chemical shift (ppm)} = \frac{f_R - f_P}{f_N}(10^6) \quad (24-4)$$

For hydrogen, with a gyromagnetic ratio of 42.6 MHz/tesla, a magnetic field strength of 1.5 tesla yields a nominal resonance frequency of

$$\begin{aligned} f &= \gamma B \\ &= (42.6 \text{ MHz/T})(1.5\text{T}) \\ &= 63.9 \text{ MHz} \\ &= f_N \end{aligned}$$

Rewriting Eq. (25-4), we have

$$\begin{aligned} f_R - f_P = \text{Frequency shift} &= \frac{(\text{Chemical shift})(f_N)}{10^6} \\ &= \frac{(3)(63.9)}{10^6} \\ &= 0.000192 \text{ MHz} \\ &= 192 \text{ Hz} \end{aligned}$$

**MARGIN FIGURE 24-23**

Left: Conventional spin echo TR = 2000, TE = 30 shows only subtle difference in signal intensity between left and right hepatic lobes.

Right: Chemical shift image (phase contrast “opposed phase” technique with same TR, TE as image on the left) shows high-intensity abnormalities consistent with metastatic cancer. Overall hepatic signal intensity is decreased due to diffuse fatty infiltration. The metastases do not contain MR observable fat, and their signal intensity relative to surrounding liver is increased. (From Stark, D. D., and Bradley, W. G. *Magnetic Resonance Imaging*. St. Louis, Mosby-Year Book, 1988, p. 240. Used with permission.)

Other factors, such as (a) variations in magnetic susceptibility of biologic materials within the body and (b) the presence of a strong water signal, render detection of small differences in resonance frequency problematic.

One of the main barriers to effective chemical shift imaging is that conventional 2DFT, the most widely used technique for spatial encoding of the MR signal, already uses frequency encoding for spatial location. Therefore, assignment of small variations in resonance frequency requires an alternative technique. Several options for signal localization have been considered.⁸⁰ One technique involves doing away with the frequency-encoding or read gradient and using only phase-encoding gradients stepped in both the x and y directions (Margin Figure 24-8). The obvious drawback of this technique is the need for $N \times N$ separate signal acquisitions, where N is the number of picture elements. Another CSI technique involves standard pulsing schemes for imaging with additional RF pulses.^{81–83} The pulses for imaging are applied in the presence of a magnetic field gradient to allow spatial selection. Other pulses, applied in the absence of a magnetic field gradient, are used to excite certain lines in the spectrum of interest. To eliminate one or more lines from the spectrum that may obscure lines of interest, a “presaturation pulse” may be administered at the appropriate frequency to excite the unwanted line, followed by a dephasing gradient pulse to remove it from the image.^{82,83}

A chemical shift image is shown in Margin Figure 24-23. At the present time, clinical applications of CSI have involved separation of fat and water components because they occur as relatively separable strong spectral lines. There is increasing interest in using spectral information from the phosphorous contained in various metabolites,⁸⁴ fluorine in blood-substitute compounds,⁸⁵ and intracellular and extracellular sodium.⁸⁰ Continued evolution of CSI shows considerable promise for the future.⁸⁶

PROBLEMS

*24-1. A spin-echo pulse sequence has a TE of 100 msec. A new pulse sequence begins 800 msec after the MR signal or echo is received. Three signal averages and 256 phase encoding steps are used. Find the total image acquisition time.

*24-2. Assume that relaxation times for gray matter and white matter in the brain are as follows:

	T1	T2
Gray matter	800	100
White matter	700	90

In which one of the following pulse sequences would the white matter appear lighter than gray matter in the image?

- TR = 750 msec, TE = 20 msec
- TR = 2000 msec, TE = 750 msec
- TR = 2000 msec, TE = 95 msec
- TR = 1500 msec, TE = 200 msec

*24-3. The frequency shift for a spectral peak on an MR spectroscopy system is 200 Hz at a static magnetic field strength of 5.0 tesla. What is the frequency shift of the same peak at a static magnetic field strength of 7.0 tesla?

24-4. Describe the mechanism of contrast enhancement when gadolinium is used as a contrast agent for MRI.

24-5. Describe three motion suppression techniques for MRI that involve acquiring the image in a shorter time (i.e., “fast scan” techniques).

*For problems marked with an asterisk, answers are provided on p. 493.

SUMMARY

- Immediately after the application of the first 90-degree pulse in a sequence, the maximum MR signal is produced. However, free induction decay causes the signal to disappear rapidly.
- The spin-echo technique allows recovery of an MR signal at a preselected time (TE) and removes the effects of T2* relaxation.
- In spin-echo pulse sequences, the choice of TR primarily affects T1 contrast and the choice of TE primarily affects T2 contrast in the image.

- Materials having greater spin density tend to have a stronger MR signal for all values of TR and TE.
- Gradient magnetic fields are used to isolate a slice and then allow frequency and phase encoding of the signal.
- In two-dimensional Fourier transform image acquisition, the resulting image is a mapping of MR signal as a function of phase and frequency.
- Material may be classified according to their magnetic properties as
 - Diamagnetic
 - Paramagnetic
 - Ferromagnetic
- Motion suppression techniques include
 - Respiratory gating
 - Respiratory-ordered phase-encoding
 - Gradient moment nulling
 - Gradient echo techniques
 - Half-Fourier imaging
- The physiological basis of functional MRI (fMRI) is that neuronal activity in localized regions of the brain changes the level of blood oxygenation in those regions. Deoxyhemoglobin is paramagnetic, but oxyhemoglobin is diamagnetic. The increase in oxyhemoglobin concentration in capillaries and veins that follows neuronal activation alters T2 and T2* contrast in the region of neuronal activation.
- Spectroscopy is an analysis of the small (parts per million) shift in resonance frequency of nuclei as a result of surrounding chemical bonds.
- The chemical shift between water and lipids is large enough to affect frequency encoding of the MR signal and can affect the appearance of an MR image or be used as the basis of chemical shift imaging.

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MAGNETIC RESONANCE IMAGING: INSTRUMENTATION, BIOEFFECTS, AND SITE PLANNING

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OBJECTIVES

After completing this chapter, the reader should be able to:

- List the components of an MRI system, state their principle of operation, and describe their contribution to the system.
- List some commonly encountered MRI artifacts, their causes, and their remedies.
- List typical quality assurance tests and state the frequency with which they should be performed.
- Give examples of bioeffects and explain the biophysical reason for each bioeffect.
- Discuss the factors that must be considered in MRI site planning.

MAIN SYSTEM MAGNET

In magnetic resonance imaging (MRI) and spectroscopy, the static magnetic field is supplied by the main system (static) magnet. This magnetic field defines the axis about which protons precess. The static field direction is also the reference axis for the nutation angle of the protons following the application of radio-frequency (RF) pulses. A range of field strengths is used for imaging, from millitesla up to 10 tesla in some research systems, with the bulk of clinical imaging conducted in the range of 0.1 to 3.0 tesla. Currently, it is not clear whether there is an optimum field strength for certain imaging situations. Some say that “more field strength is better” if the engineering challenges presented by high-field strength systems can be overcome. It is true that the RF signal increases with field strength.¹ T1 also increases with field strength, however, and the magnitude of T1 influences the selection of parameters for the pulsing scheme. These, in turn, influence image-degrading factors such as noise and patient motion. Furthermore, it is the relative contrast between two tissues, as compared with noise, that is of interest in imaging. For a given field strength, therefore, the optimization of pulse parameters depends upon the tissues of interest.²

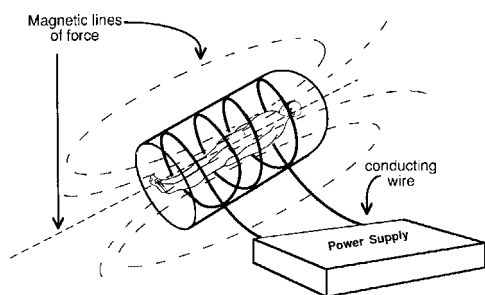
Other challenges associated with the use of higher field strengths for MRI include difficulties in maintaining magnetic field homogeneity, increased RF heating, and chemical shift artifacts. The Larmor equation calls for higher RF frequencies as field strength is increased. Higher frequencies are more readily absorbed, which causes more RF heating of the patient. Chemical shift artifacts occur because higher field strengths shift the resonance frequencies of fat and water molecules. Because frequency encoding is used for spatial localization, the position of fat layers may be shifted in the image. One might infer from these challenges that lower field strengths will continue over the foreseeable future to be useful in magnetic resonance imaging.

There are two types of main system magnets: electromagnets and permanent magnets.

Electromagnets

An electromagnet is formed by twisting conducting wire in the shape of a coil or solenoid. Any wire that carries an electric current is surrounded by a magnetic field. When the conducting wire is twisted into the shape of a coil, the magnetic field from the coil loops reinforce each other to produce a magnetic field that is strongest in the coil center. Magnetic field “lines” are invisible lines in space that define the direction of force of the magnetic field upon a ferromagnetic object such as iron. The magnetic field lines at the center of a solenoidal electromagnet lie along the axis of the coil (Margin Figure 25-1).

Two types of electromagnets, resistive and superconductive, are used in MRI. The field strength of an electromagnet is limited by the amount of current that may be carried by the conducting wire. Some portion of the energy of an electric current is dissipated as heat in the wire because the conducting material resists the flow of electrons. Because of this heat, resistive-type MRI magnets must be water-cooled.



MARGIN FIGURE 25-1

A solenoidal electromagnet produces magnetic lines of force that are concentrated along the axis of the solenoid or coil.

Attempting to force high currents through the wire produces excessive heat in the wire. For this reason, the maximum field attainable with a resistive MRI magnet is usually below 0.5 tesla. Power usage is also an issue; a typical resistive MRI system requires approximately 30 kW of power, the equivalent of 300 100-W light bulbs.

To achieve higher field strengths in magnets that are large enough for imaging human patients, conducting wire must be used in which there is no resistance. This is the property of superconductivity described in Chapter 2. Superconductive magnets allow higher currents and therefore higher field strengths. To maintain superconductivity, conductors must be kept near the temperature at which the element helium is a liquid (-269°C). This temperature is near absolute zero (0°K , or -273°C). Liquid helium is circulated around the superconducting wire in an insulating chamber called a dewar. A second insulating chamber, containing liquid nitrogen at -196°C , is used to help maintain the helium in its liquid state.

Clinical MRI using superconducting magnets is now routinely performed at magnetic field strengths of up to 2 tesla. Research with systems having bore sizes sufficiently large for humans is performed at up to 9 tesla. Small-bore research systems for animal imaging and spectroscopy exist at field strengths up to 10 tesla. There is hope that recent advances in the fabrication of ceramics that exhibit superconductivity at elevated temperatures may permit significant advances in magnetic resonance (MR) magnet technology in the not-too-distant future.

Permanent Magnets

Permanent magnets are composed of ferromagnetic materials that maintain their magnetic properties with no additional input of energy. Large assemblies of ferromagnetic materials for clinical MRI systems have field strengths of up to 0.3 tesla. In such a system the patient is placed between the poles of the permanent magnet with the direction of magnetic field lines perpendicular to the axis of the patient (Margin Figure 25-2).

GRADIENT MAGNETIC FIELDS

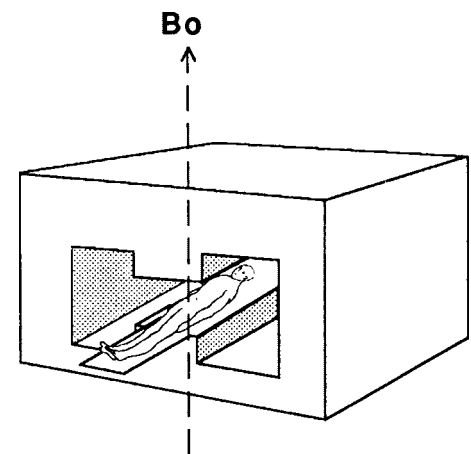
The gradients required for spatial encoding of the MR signal are provided by gradient electromagnetic coils. The magnetic field generated by a gradient coil is substantially smaller than the field of the main system magnet. A typical rate of change of field strength is several gauss per centimeter over a distance of 100 cm. By applying the current in opposite directions to two loops on either side of the patient, magnetic fields are generated to provide a range of field strengths over the patient. These conducting loops are oriented to provide variations in magnetic field intensity along three orthogonal axes of the patient. An illustration of the orientation of the coils is shown in Margin Figure 25-3.

RADIO-FREQUENCY COILS

RF coils send and receive radio waves in a fashion similar to a radio antenna. The term “coil” is used rather than “antenna” when the source and receiver are very close, as is the case in MRI. When pulses are sent into the patient, the coil is the source, and the patient is the receiver. When the MR signal is read out, the patient is the source, and the coil is the receiver. In either case, the source and receiver are within a few coil diameters of each other.

Sending and Receiving Coils

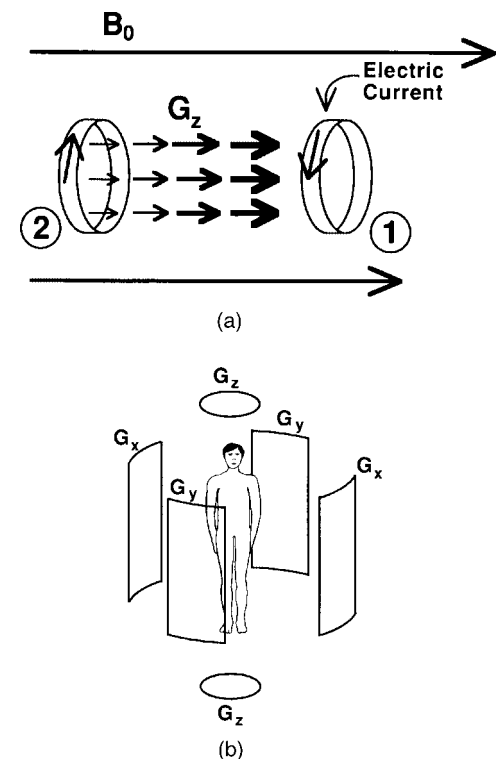
The sending coil of an MRI system is usually built into the bore of the magnet (Margin Figure 25-4). Precise positioning of the patient within the sending coil is necessary



MARGIN FIGURE 25-2

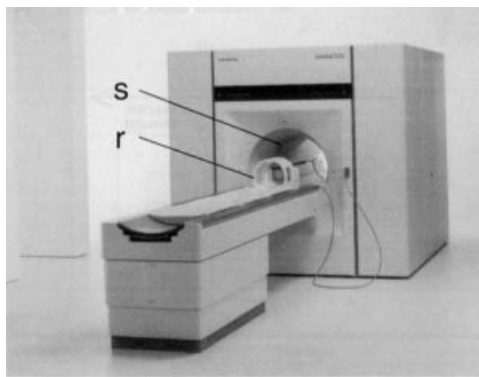
Diagram of a permanent magnet MRI gantry. The orientation of the patient is shown in the static magnetic field (B_0) produced by the poles of the magnet.

1 gauss is equal to 0.1 millitesla.

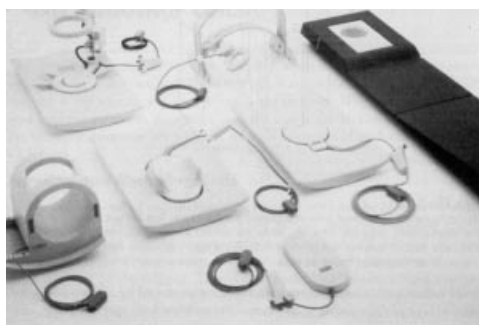


MARGIN FIGURE 25-3

A: Configuration of gradient coils in opposing current loops. The current in coil 1 augments the main-system magnetic field, B_0 , while the current in coil 2 opposes the main-system magnetic field. The result is a gradient field, G_z , a field that varies with position along the z -axis. **B:** An illustration of the six coils that result in three gradient fields, G_x , G_y , G_z .



(a)



(b)

MARGIN FIGURE 25-4

A: The radio-frequency (RF) coil for sending is usually located within the bore of the magnet, while the RF coil for receiving is located on the patient table. **B:** An assortment of RF “surface coils” for imaging various body parts. (Courtesy of Siemens Medical Systems.)

to ensure that the anatomy of interest appears in the image. In most imaging units it is difficult to visualize patient position after the patient has entered the bore of the magnet. Therefore, a positioning system is used that involves the use of external markers or a patient table indexing scale.

Under some circumstances, the use of separate sending and receiving coils may be eliminated. The same coil may send and receive radio waves in the same fashion as radio antennas may be both senders and receivers. There is some interest in eliminating the in-bore sending coil to provide more room in the bore to help alleviate claustrophobic effects in patients. Alternately, deletion of the in-bore coil would allow a reduction in bore size that could resolve some of the design problems inherent in MRI magnet construction. The RF pulse produced in a small surface coil is less uniform than that provided by an in-bore solenoidal sending coil. It is possible to overcome this limitation with special pulsing techniques.³

Factors Affecting the Performance of Receiver Coils

Margin Figure 25-4 shows an assortment of receiver coils. The shape of the coil may be influenced by the body part being imaged. For small samples, a solenoidal coil completely surrounding the sample is often used.⁴ For many parts of the human body, however, a surface coil is more practical. A surface coil is essentially a loop of conducting material, such as copper tubing, that is placed adjacent to the body part being imaged. The loop may form various shapes and be bent slightly to conform to a body part such as a joint.

A rule of thumb for surface coils is that the sensitivity decreases appreciably beyond a distance equal to the diameter of the coil.⁵ Thus, positioning of the coil is an important determinant of performance. Surface coils, for example, are often positioned on the patient prior to entry of the patient into the magnet bore.

A fundamental limitation in the sensitivity of a receiver coil is the presence of spurious electrical signals (electronic noise) that do not contribute diagnostic information to the MR signal. A major source of noise is the movement of electrically charged molecules such as electrolytes within the patient. Other sources of noise produced in the coil itself are usually of smaller magnitude than patient noise.⁶ Improvements in imaging performance are usually more achievable through increased signal rather than decreased noise.

Maximization of signal from a receiver coil requires “tuning” of the coil to a resonance frequency. The resonance frequency for a coil is the frequency at which transfer of energy from the patient to the coil is most efficient. Just as an organ pipe or a bell has a natural frequency of vibration (musical tone), an electronic component such as a coil has a natural electrical frequency. Operation of a coil at frequencies that are “off resonance” results in a reduction in signal strength.

The resonance frequency of a coil is determined by the geometry of the coil, including its length, diameter, number of loops, and so on. The resonance frequency may be adjusted by adding or subtracting capacitance, the ability of a circuit to store electrical charge. The resonance frequency of a coil changes when the coil is “loaded”—that is, when a patient is inserted into the magnet. The frequency of the coil may then have to be readjusted by changing the setting of variable or “tuning” capacitors. This adjustment may be an automatic part of signal optimization in the system, or the operator may have to make manual adjustments.

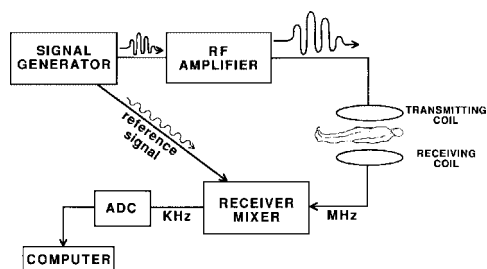
**MARGIN FIGURE 25-5**

Diagram of the main electronic components of an MRI system. The functions of the various components are described in the text.

ELECTRONIC COMPONENTS

The operation of an MRI system requires the use of several standard electronic components (Margin Figure 25-5). The transmitted radio waves originate in a device known as a signal generator. The signal generator's task is to produce a signal at a precise frequency and desired shape. The magnitude of the signal is then increased by an RF amplifier.

The receiver electronics include a mixer that combines the received signal (MR signal obtained from the patient by the receiver coil) with a reference signal at or near the Larmor frequency. The effect of mixing the signals is to subtract out the Larmor frequency, leaving only the small shifts in frequency. These frequency shifts are used for spatially encoding the MR signal. After mixing, a low-frequency (kilohertz instead of megahertz) MR signal remains. The purpose of mixing is to provide a low-frequency signal that can be digitized more precisely than a high-frequency signal. In a given cycle of the signal, the analog-to-digital converter (ADC) has more time to divide the signal into successive digital values to yield greater precision.

■ COMPUTER

As signals from the ADC are acquired, they are transferred to a temporary storage location in the computer as they are acquired. This storage area, called the buffer, allows the ADC to work at maximum speed. Additional memory, such as random access memory (RAM) in the computer, must be of sufficient size to permit storage of all pulse sequence software instructions. These instructions include the acquisition scheme [e.g., two-dimensional Fourier transform (2DFT)], number of slices, number of phase-encoding steps, and number of signal averages. An array processor is required for rapid reconstruction of images. Because the array processor must be able to operate on an entire data set, RAM must include space for a least several images. External storage of images may then be accomplished on magnetic tape or disk. Increasingly, both external and long-term storage are provided by optical disk systems.

■ ARTIFACTS

Artifacts in MR images may be grouped according to the system components most responsible for their occurrence (e.g., static magnetic field, gradient magnetic field, and RF artifacts). Other artifacts include motion artifacts (including flow), chemical shift, and foreign object artifacts.

Foreign Objects

Most tissues in the body are diamagnetic and repel magnetic fields slightly. If paramagnetic or ferromagnetic objects, which attract magnetic fields, are present within tissue, they distort the magnetic field (Margin Figure 25-6) and may affect spatial encoding. For example, a ferromagnetic object produces a strong local magnetic field, as much as 100 times the static field of the magnet in its vicinity. This distortion of the magnetic field creates a signal void (dark region) surrounded by a white border at the location of the artifact. The extent of the distortion depends upon the size and composition of the object. Some common foreign objects that produce susceptibility artifacts include surgical clips, prosthetics, and shrapnel.

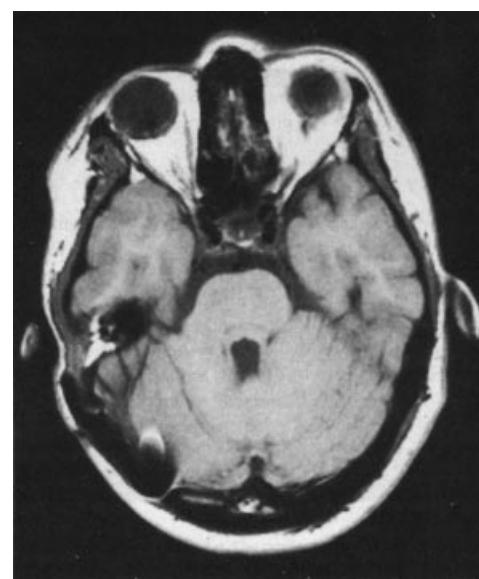
Magnetic Field

Factors other than foreign objects may produce artifacts by causing abrupt changes in the static magnetic field across an image. For example, shim coils may be improperly adjusted, or inhomogeneities may arise at the edges of especially large fields of view. These effects cause distortion of the image due to disruption of spatial encoding.

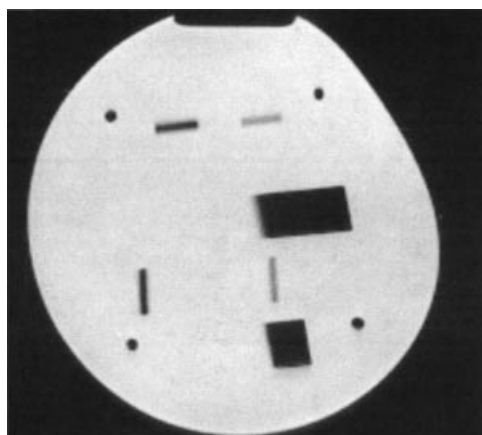
Gradient Field

The usual spatial-encoding schemes assume that the gradient magnetic fields are linear. If nonlinearities exist along any of the gradient axes (slice, frequency, or

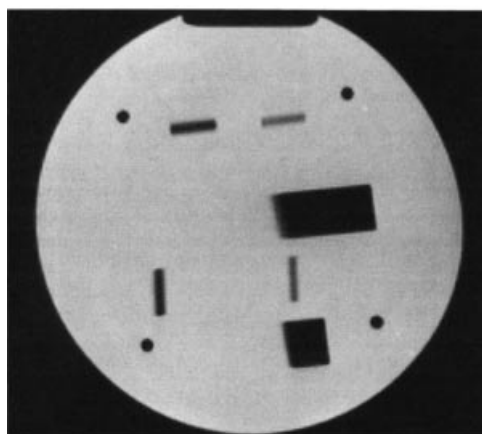
Foreign object artifacts are sometimes referred to as “susceptibility” artifacts.



MARGIN FIGURE 25-6
Radio-frequency (RF) artifact caused by a metal object in the field of view.



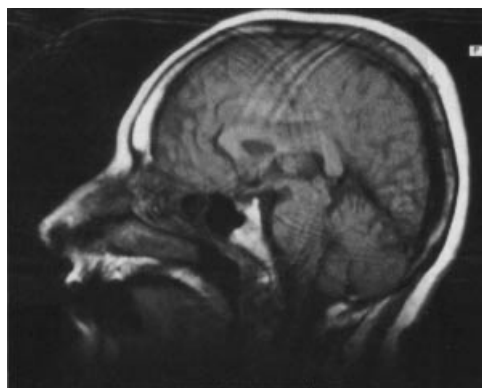
(a)



(b)

MARGIN FIGURE 25-7

Artifact caused by gradient nonlinearity. The artifact may be alleviated by switching the phase-encoding direction to coincide with the direction in which the gradient is most nonlinear.

**MARGIN FIGURE 25-8**

Artifact caused by patient motion. The artifact is propagated along the direction in which phase encoding is performed.

phase-encoding directions), then variations in signal intensity will occur (Margin Figure 25-7). The phase-encoding process is the least sensitive of the three to small variations in linearity. Therefore, a standard technique for reducing the effect of gradient field nonlinearities is to choose the phase-encoding direction to coincide with the gradient direction with poorest linearity.

Radio Frequency

Radio waves from an outside source can interfere with signals received from the patient and be misinterpreted as pictorial information. These RF artifacts may appear as a line or band across the image in a direction perpendicular to the frequency-encoding direction. They represent RF “noise” as a single frequency or narrow range of frequencies. Artifacts may also affect the image in a more general but obviously nonphysiologic fashion such as a “herring bone” pattern.

Some RF-related problems in MR images are not caused by outside interference but instead by problems with internal components of the system such as malfunction of the RF transmitter, poor electrical connections, or failure of circuits associated with the receiver coil.

Motion

In MRI, patient motion causes artifacts that project through the image in the direction of phase encoding (Margin Figure 25-8). Phase encoding is usually performed over a number of phase-encoding steps, whereas frequency encoding takes place during a small number of pulse repetition periods. For this reason, spatial localization of signals in the phase-encoding direction is more easily disrupted than is localization along the frequency axis. Common sources of motion artifacts are gross patient movement, respiratory motion, cardiac motion, and flow. Flow artifacts may appear as decreased, increased, or variable changes in image brightness. Rapid flow may appear as decreased intensity because spins entering the slice during signal readout were not present during slice selection and do not provide a coherent signal. Less rapid flow may actually result in a signal increase because spins that were not saturated (nutated into the transverse plane) may be saturated by a subsequent pulse. Thus, high signal value in flowing material does not necessarily signify the effects of relaxation processes as would be the case for stationary material.

The flow phenomenon of enhanced signal during even-numbered echoes has been widely recognized.⁷ In this artifact, seen in vessels exhibiting slow laminar flow, spins are not completely rephased by the initial 180-degree pulse. A subsequent 180-degree pulse completes rephasing, but the next is incomplete and so on. Much later echoes may fail to yield any signal as the spins leave the slice of interest.

Chemical Shift

Small differences in resonance frequency among different chemical environments of protons are not usually revealed in MR imaging. Chemical shifts produced by the screening effects of the electron “clouds” of nearby chemical bonds are usually on the order of tens of parts per million. Spatial localization using magnetic field gradients places signals in the appropriate pixels as a reflection of changes in field strength on the order of several parts per thousand—that is, one pixel out of the matrix size of 256 or 512. Thus chemical shifts do not usually affect the MR imaging process directly. An exception to this occurs when lipid-rich tissues are adjacent to tissues with high water content (Margin Figure 25-9). The lipid/water chemical shift is among the largest encountered in normal body tissues and may be detected as an artifact along the frequency-encoding direction. A region of increased signal intensity appears at one margin of the object, while a region of decreased signal appears at the opposite margin. The signal from lipid is shifted in the direction of decreasing magnetic field,